MUSCLE OS

PILLAR 2

Training & Programming

the complete guide to strength and physique development

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Domain 1: Training Foundations

1. Mechanical Tension, Metabolic Stress & Muscle Damage

1.1 The Three Mechanisms of Hypertrophy

Skeletal muscle hypertrophy is driven by three primary mechanisms that operate synergistically: mechanical tension, metabolic stress, and muscle damage. While each contributes to the hypertrophic response, mechanical tension is universally accepted as the primary driver. Understanding how these mechanisms interact allows the coach to design training programmes that maximise adaptive signalling while managing recovery.

Key Distinction

Mechanical tension is **necessary and sufficient** for hypertrophy. Metabolic stress and muscle damage are **adjuvant** mechanisms that can augment the response but cannot independently produce substantial hypertrophy in the absence of mechanical tension.

1.2 Mechanical Tension

Mechanical tension refers to the force produced by a muscle during contraction, combined with the passive tension generated through muscle stretch. The total tension stimulus is the product of force production and muscle length at peak tension. This is why exercises that load a muscle in its lengthened position (deep stretch under load) produce superior hypertrophic responses — they maximise both active force and passive tension simultaneously. The mechanotransduction pathway converts this mechanical signal into a biochemical response: integrins, focal adhesion kinase (FAK), and the YAP/TAZ pathway transduce sarcolemmal strain into mTOR activation and subsequent protein synthesis.

1.3 Metabolic Stress

Metabolic stress refers to the accumulation of metabolites — lactate, inorganic phosphate (Pi), hydrogen ions (H+), and reactive oxygen species — during high-repetition, moderate-load, short-rest training. These metabolites contribute to hypertrophy through several pathways: cell swelling (volume expansion of the muscle fibre activates anabolic signalling via the osmosensitive transcription factor TonEBP/NFAT5), hypoxia-induced transcription factors (HIF-1a upregulates VEGF, supporting capillary density and nutrient delivery), and metabolite-mediated growth factor release (lactate has been shown to stimulate muscle stem cell activity independent of pH changes).

1.4 Muscle Damage

Exercise-induced muscle damage involves Z-disc disruption, sarcomere disorganisation, and the resulting inflammatory response. Satellite cell activation is the primary repair mechanism: quiescent satellite cells are activated by damage signals (HGF release from the extracellular matrix) and proliferate to donate nuclei to existing myofibres, increasing the myonuclear domain and supporting further hypertrophy. The inflammatory response includes neutrophil infiltration, macrophage polarisation (M1 to M2 transition), and release of cytokines (IL-6, IL-10) that coordinate repair and remodelling. While some damage is necessary for adaptation, excessive damage impairs recovery and interferes with subsequent training quality.

1.5 Synergistic Operation

Table 1.1 The Three Hypertrophy Mechanisms

MECHANISM	STIMULUS	PRIMARY VARIABLE	KEY EVIDENCE
Mechanical tension	Force production x muscle stretch	Load, full ROM, maximally stretched position under load	Schoenfeld (2010); Hornberger & Chien (2006)
Metabolic stress	Metabolite accumulation (lactate, H+, Pi)	Rep range (8-15+), rest intervals (30-90 s), continuous tension	Schoenfeld (2013); Burd et al. (2012)
Muscle damage	Z-disc disruption, eccentric loading	Novel exercises, eccentric emphasis, high volume	Clarkson & Hubal (2002); Toigo & Boutellier (2006)

CHAPTER SUMMARY

Mechanical tension is the primary hypertrophy driver, with metabolic stress and muscle damage playing adjuvant roles. Exercises that load the muscle in its lengthened position maximise the tension stimulus. Cell swelling from metabolite accumulation and satellite cell activation from muscle damage contribute to but cannot substitute for mechanical tension. A well-designed programme exploits all three mechanisms through appropriate exercise selection, loading parameters, and rest intervals.

◈ MAKE THIS WORK FOR YOU

Your training should prioritise exercises that maximise mechanical tension in the stretched position. For most people, this means choosing movements that allow a full range of motion under load — deep squats, Romanian deadlifts from the floor, dumbbell presses with a full stretch at the bottom. The simplest change you can make is to extend your ROM in the lengthened position while keeping the load moderate enough to maintain control.

If hypertrophy is your goal: Structure your session so at least one exercise per muscle group loads the stretched position significantly. For chest: dumbbell flyes or incline dumbbell press before flat barbell press. For quads: hack squat or leg press (deep knee flexion) before leg extension.

If you are a beginner or returning from a layoff: Mechanical tension alone, applied consistently through full ROM, will drive most of your early gains without excessive soreness. Keep early sessions moderate in volume (8–12 working sets per muscle per week) and focus on exercise quality rather than chase extreme metabolite burn.

If you are an advanced trainee (3+ years): You may need to exploit adjuvant mechanisms more deliberately — higher metabolite training (moderate loads, 30–60 s rests, extended sets) or occasional eccentric overload — to continue progressing. The advanced lifter's challenge is not stimulating growth but doing so while managing recovery.

2. Progressive Overload — Definitions and Methods

2.1 Defining Progressive Overload

Progressive overload is the systematic increase in training demand over time that forces the body to adapt beyond its current capacity. Without progressive overload, adaptation plateaus; the stimulus that once drove progress becomes maintenance. The principle applies across all training goals: strength, hypertrophy, endurance, and power. The art of coaching lies in selecting the appropriate overload method and progressing it at the right rate for the individual.

2.2 Methods of Progressive Overload

Table 2.1 Progressive Overload Methods

METHOD	HOW TO APPLY	WHEN TO PROGRESS
Load increase	Add 2.5–5 kg (upper body) or 5–10 kg (lower body) to the working weight	When target reps are achieved with good form at current load
Volume increase	Add 1–2 sets per exercise or add an additional exercise per muscle group	When plateauing in a given rep range with stable recovery
Frequency increase	Add an additional training day or increase per-muscle frequency	When volume per session exceeds recovery capacity (session RPE > 8)
Density increase	Reduce rest intervals by 15–30 seconds while maintaining load and volume	When inter-set recovery feels excessive for the goal (hypertrophy: rest below 90 s)
Tempo manipulation	Slow the eccentric phase (3–5 seconds) or add an isometric pause at the stretched position	When time under tension needs to increase without changing load
Range of motion increase	Increase depth, stretch, or end-range loading	When mobility allows deeper ROM without form breakdown

2.3 Double Progression Method

The double progression method is the most practical approach for hypertrophy-oriented training. It involves two phases within a given rep range: **Phase 1** — once the target rep range is achieved with good form, add weight. **Phase 2** — after adding weight, the reps will drop. Build back up to the top of the rep range over subsequent sessions. For example: target 8–12 reps. When 12 reps are achieved with good form, increase the load by 2.5–5 kg. The next session, 8–9 reps will be achievable; build back to 12 over 2–4 weeks. This method provides a clear progression framework while allowing for normal performance fluctuation.

Progression Decision Process

1. **Goal-specific range:** Establish the target rep range for the exercise (e.g., 8–12 for hypertrophy, 3–5 for strength).
2. **Load selection:** Choose a load that places you at the lower end of the range (e.g., 8 reps).
3. **Rep accumulation:** Over subsequent sessions, build reps within that range while keeping the load constant.
4. **Load progression trigger:** When the upper rep boundary is reached (e.g., 12 reps) with controlled form, increase load by 2.5–5%.
5. **Cycle repeat:** The increased load will drop you to the lower end. Repeat the accumulation phase.
6. **Plateau intervention:** If unable to progress load or reps within 3–4 weeks, assess: insufficient volume? Poor recovery? Exercise selection issue?

2.4 Rate of Progression Expectations

Novice lifters can progress load session-to-session (linear progression). Intermediate lifters progress week-to-week or using double progression. Advanced lifters may require mesocycle-level progression (4–8 weeks) to see measurable load increases. The rate of progression slows as training age increases, reflecting the diminishing returns curve of the adaptive response.

CHAPTER SUMMARY

Progressive overload is the non-negotiable driver of adaptation. Six methods are available: load increase, volume increase, frequency increase, density increase, tempo manipulation, and ROM increase. Double progression is the most practical framework for hypertrophy training. Rate of progression slows with training age: novice lifters progress session-to-session, intermediates within-weeks, and advanced lifters mesocycle-to-mesocycle.

◈ MAKE THIS WORK FOR YOU

Your rate of progression should match your training age, not your ambition. A novice can add weight every session because the nervous system is learning to coordinate and the muscles are adapting rapidly. An advanced lifter may need 4–8 weeks to add 5 kg to a lift — and that is normal progress, not a plateau. The most common programming mistake is trying to progress faster than your training age allows.

If you are a novice (under 1 year of consistent training): Use simple linear progression — add weight to the bar every session if your target reps are achieved. If you complete 3×8 with good form, add 2.5–5 kg next session. When you can no longer add weight every session, switch to weekly progression (add weight each week) or the double progression method.

If you are intermediate (1–3 years): Double progression is your most reliable framework. Pick a rep range (e.g., 8–12), build reps to the top of the range, then add weight. This naturally accommodates the ups and downs of daily performance. A 2.5–5 kg increase every 2–4 weeks per major lift is a good cadence.

If you are advanced (3+ years): Mesocycle-level progression (4–8 week blocks) is realistic. Use block periodisation: one block focusing on volume accumulation (higher reps, moderate loads), then a block focusing on intensity (lower reps, heavier loads). If you are adding 2.5–5 kg to your squat every 4–8 weeks, you are still progressing well.

3. Volume Landmarks (MEV, MAV, MRV) and Volume Progression

3.1 The Individual Response Curve

Training volume — typically quantified as the number of hard sets per muscle group per week — follows an individualised dose-response relationship. The response curve has three critical landmarks: the Minimum Effective Volume (MEV), the Maximum Adaptive Volume (MAV), and the Maximum Recoverable Volume (MRV). These landmarks shift with training age, recovery capacity, muscle group size, and anabolic context (surplus vs. deficit). The coach's primary task is to locate the client's MAV through systematic volume adjustment.

> **Volume Landmarks Defined**
>
> **MEV (Minimum Effective Volume):** The smallest weekly set volume that produces a measurable hypertrophic response. Training below MEV results in maintenance, not growth.
>
> **MAV (Maximum Adaptive Volume):** The volume at which the hypertrophic response is maximised per unit of recovery capacity. Training at MAV produces the best growth-to-fatigue ratio.
>
> **MRV (Maximum Recoverable Volume):** The highest weekly set volume that can be recovered from without performance decline, illness, or injury. Training at or above MRV without strategic deloading leads to overtraining.

3.2 Volume Landmarks by Muscle Group

Table 3.1 Volume Landmarks by Muscle Group (Sets per Week)

MUSCLE GROUP	MEV (SETS/WK)	MAV (SETS/WK)	MRV (SETS/WK)
Quadriceps	6-8	12-16	20-25
Hamstrings	4-6	8-12	16-20
Glutes	6-8	12-16	20-25
Chest (pectorals)	6-8	10-14	18-22
Back (lats, rhomboids, traps)	8-12	14-20	25-30
Shoulders (deltoids)	4-6	8-12	16-20
Biceps	4-6	8-12	16-20

Triceps	4-6	8-12	16-20
Calves	4-6	8-12	16-20
Abs / core	4-6	8-12	16-18

3.3 Finding MAV: The Systematic Adjustment Protocol

How to Find Your MAV

1. **Start at MEV:** Begin with the minimum effective volume for the muscle group.
2. **Assess after 3–4 weeks:** Measure progress (weight on bar, reps, measurements). If progressing, you are at or below MAV.
3. **Add 1–2 sets per muscle group:** Increase volume by one set per exercise or add an exercise. Maintain for 3–4 weeks.
4. **Reassess:** Compare progress rate against the previous block.
5. **Continue adding until:** Progress plateaus, recovery declines, or systemic fatigue accumulates faster than adaptation.
6. **Back off to the previous volume:** The volume just below the plateau is your MAV.

3.4 Volume Progression Across a Training Block

A typical mesocycle (6–12 weeks) might start slightly below MAV and progress toward or slightly above MAV by the final weeks. The first 2–3 weeks serve as a volume acclimation phase. The middle 4–6 weeks operate at or near MAV. The final 1–2 weeks may approach MRV before a planned deload. This phased approach allows the athlete to accumulate volume without prematurely overwhelming recovery capacity.

CHAPTER SUMMARY

Training volume follows an individualised dose-response curve defined by MEV, MAV, and MRV. Start near MEV and systematically add sets every 3–4 weeks until progress plateaus or recovery declines. MAV varies by muscle group, training age, and anabolic context. A typical mesocycle progressively increases volume from slightly below MAV to slightly above, followed by a deload. Volume ranges provided in Table 3.1 are starting estimates — individual variation is substantial and must be empirically determined.

◈ MAKE THIS WORK FOR YOU

Volume landmarks are individual, not universal. The ranges in Table 3.1 are population estimates — your MEV, MAV, and MRV depend on your training age, recovery capacity, muscle fibre type distribution, sleep quality, stress levels, and nutritional status. Two identical programmes can produce different results for two different people because their volume landmarks differ by 30–50%.

Finding your starting volume: For each major muscle group, start at the lower end of the MEV range. After 3–4 weeks, assess: are you making progress (strength increasing, measurements trending up)? If yes, you are at or below MAV. Add 1–2 sets per muscle group and re-assess after another 3–4 weeks. Continue this process until progress stalls — the volume just below the stall point is your current MAV.

If you are a beginner: Your MEV and MAV are lower than you think. Starting at 8–10 sets per muscle group per week is usually sufficient for progress. More volume is not more better — the beginner nervous system and musculature respond to relatively low stimulus. Focus on exercise quality and progressive overload rather than adding sets prematurely.

If you are training in a caloric deficit: Your MRV is reduced (by 20–30% typically) because the energy required to recover from training is limited. Do not try to maintain the same volume you used in a surplus. Reducing volume by 1–3 sets per muscle group while maintaining intensity (load) preserves muscle better than high-volume, low-intensity training in a deficit.

4. Intensity, Proximity to Failure & RPE/RIR

4.1 Two Meanings of Intensity

Intensity has two distinct but related meanings in resistance training. **Load-based intensity** refers to the percentage of your one-repetition maximum (%1RM) used for an exercise. **Effort-based intensity** refers to proximity to momentary muscular failure, quantified by RPE (Rating of Perceived Exertion) or RIR (Reps In Reserve). These two concepts are independent: a lifter could perform a set at 70% 1RM (moderate load intensity) but take it to RPE 10 (maximal effort intensity), or perform a set at 90% 1RM but stop at RPE 7 (high load intensity, low effort intensity).

4.2 The RPE/RIR Scale

Table 4.1 RPE/RIR Reference Scale

RPE	RIR	DESCRIPTION
10	0	Momentary muscular failure; cannot complete another rep with good form
9.5	0-1	Could possibly complete 0-1 more rep, but form would break down
9	1	One rep left in the tank; could complete one more rep with good form
8.5	1-2	Between 1 and 2 reps remaining
8	2	Two reps left in the tank; comfortable effort
7.5	2-3	Between 2 and 3 reps remaining
7	3	Three reps left; moving weight feels smooth and controlled
6.5	3-4	Approaching challenging but still very controlled
6	4+	Warm-up / technique work; no significant fatigue

4.3 Optimal Proximity to Failure by Goal

Table 4.2 Proximity to Failure by Training Goal

GOAL	OPTIMAL RPE	RIR	FREQUENCY OF FAILURE EXPOSURE
Maximal strength (1-5 RM)	9-10	0-1	Occasional (1-2x per month per lift for testing)
Hypertrophy (6-15 reps)	7-9	1-3	Periodic (final sets of last exercise, or last sets of a mesocycle)

Muscular endurance (15+ reps)	8–10	0–2	Higher tolerance; metabolites affect perception
Power / explosive (1–3 reps)	5–7	3–5+	Never to failure (fatigue degrades power output)
Rehabilitation / return from injury	4–6	4+	Never to failure; stop well before technical breakdown

4.4 The Case for and Against Training to Failure

Training to failure (RPE 10) elevates acute muscle protein synthesis more than submaximal efforts in the same session. However, the recovery cost is disproportionately higher: failure sets produce greater systemic fatigue, longer neural recovery time, and a larger cortisol response. The evidence suggests that training to failure on every set does not produce superior hypertrophy compared to leaving 1–3 RIR, but it does increase cumulative fatigue. Strategic failure exposure — the last set of the last exercise for a muscle group, or the final week of a mesocycle — maximises the stimulus-to-fatigue ratio. For strength, failure exposure is even more costly because the central nervous system requires more time to recover from maximal-effort heavy lifts.

Practical Application

For hypertrophy, keep most working sets at RPE 7–9 (1–3 RIR). Reserve RPE 10 for the last set of each exercise, the last exercise of the session, or the final week of a mesocycle. For strength, use RPE 7–9 for volume accumulation and RPE 9–10 for intensity phases. Never train to failure during power or velocity-oriented work.

CHAPTER SUMMARY

Intensity has two meanings: load-based (%1RM) and effort-based (RPE/RIR). The RPE/RIR scale standardises proximity to failure across different rep ranges and exercises. Optimal hypertrophy RPE is 7–9 (1–3 RIR). Training to failure on every set increases fatigue disproportionately to the additional stimulus. Strategic failure exposure at the end of a session or mesocycle optimises the stimulus-to-fatigue ratio.

◈ MAKE THIS WORK FOR YOU

Learning to gauge RPE/RIR accurately is a skill that takes 4–8 weeks of consistent practice. Start by keeping a training log where you record your RPE for each set immediately after completing it. Review your log weekly: if you frequently rated sets at RPE 10 but hit your target reps, you are likely underestimating your capacity (you had more in reserve than you thought). If you consistently miss rep targets, your RPE rating may be too conservative for the load you are using.

If you are new to RPE: Use RIR (Reps In Reserve) instead of RPE — it is more intuitive. After each set, ask yourself: "How many more reps could I have done with perfect form?" If the answer is 2–3, your RIR is 2–3 (RPE 7–8). If you could not have done another rep, your RIR is 0 (RPE 10). Do not chase RPE 10 on every set — you will accumulate fatigue faster than adaptation.

If you tend to overtrain or have poor recovery: Keep most working sets at RPE 7–8 (2–3 RIR). Reserve RPE 9 for the last set of each exercise and RPE 10 for rare occasions (last week of a training block, testing a new max). The majority of your hypertrophic response occurs between RPE 7 and 9, but the recovery difference between RPE 8 and RPE 10 is disproportionately large.

If you are training for strength: Proximity to failure matters differently. Most of your strength work should be at RPE 7–9 (1–3 RIR) during accumulation phases. Only during peaking phases (2–4 weeks before a max test or meet) should you approach RPE 9–10. The nervous system recovers more slowly from heavy, near-failure sets than from volume work.

5. Exercise Selection & Biomechanics

5.1 Stability Continuum: Free Weights vs. Machines

Exercise selection lives on a stability continuum. At one end, free-weight compound movements (barbell squats, deadlifts, bench press) require significant stabiliser activation and produce the greatest systemic loading. At the other end, selectorised machines isolate target muscles with minimal stabiliser demand. Neither is inherently superior; each serves a purpose. Free weights build coordination, stabiliser strength, and transferable athletic capacity. Machines allow targeted muscle loading with precise resistance profiles, safer failure, and easier progressive overload. A well-designed programme uses both, prioritising free weights for fundamental movement patterns and machines for targeted hypertrophy work.

5.2 Range of Motion and Stretch-Mediated Hypertrophy

Full range of motion (ROM) through the lengthened position of a muscle is emerging as a critical variable for hypertrophy. The lengthened partial (the bottom portion of a lift where the muscle is under stretch) appears to produce disproportionate hypertrophic signalling compared to the shortened portion. This is mediated by the length-tension relationship: passive tension from the stretched muscle adds to the mechanical tension stimulus, and the sarcomerogenesis response (addition of sarcomeres in series) increases the muscle's adaptive capacity. Practical application: prioritise exercises that load the muscle in its fully stretched position (deep squats, incline press with stretch, Romanian deadlifts with full hamstring stretch), and ensure the stretched portion of each rep is actively controlled, not bounced through.

5.3 Biomechanical Selection Principles

Table 5.1 Biomechanical Principles for Exercise Selection

EXERCISE CATEGORY	BIOMECHANICAL FACTOR	SELECTION PRINCIPLE
Compound pressing (horizontal)	Horizontal adduction + shoulder flexion moment arm	Barbell limits ROM at the extreme stretch; dumbbells and machines allow greater pectoral stretch
Compound pressing (vertical)	Scapular plane of motion	Neutral grip (palms facing) reduces impingement risk; wide grip maximises lateral delt involvement

Compound pulling (vertical)	Latissimus dorsi moment arm changes with grip width	Wide grip targets upper lats; close grip targets lower lats and biceps
Compound pulling (horizontal)	Scapular retraction and spinal extension	Chest-supported rows eliminate lower back fatigue as a limiting factor
Squat patterns	Femur length creates moment arm differences	Longer femurs require more forward lean (low-bar) or heel elevation to maintain upright torso
Hinge patterns	Hip moment arm vs. spinal loading	RDLs target hamstring lengthened position; conventional deadlifts shift load to erectors and glutes
Single-joint isolation	Leverage curve through ROM	Select machines or cable positions that match the strength curve of the target muscle

5.4 The Principle of Specificity

The SAID principle (Specific Adaptation to Imposed Demands) states that the body adapts specifically to the stress placed upon it. For strength, this means practising the competition lift or its close variant. For hypertrophy, it means selecting exercises that maximally stretch and load the target muscle through its full ROM. Exercise selection should be goal-directed: choose the exercise that produces the best stimulus-to-risk ratio for the specific adaptation sought.

CHAPTER SUMMARY

Free weights and machines exist on a stability continuum; both are valuable tools. Full ROM through the stretched position is critical for hypertrophy. Exercise selection should be guided by biomechanical principles: moment arms, length-tension relationships, and individual anthropometry. The SAID principle dictates that exercise selection must be specific to the adaptation goal.

◈ MAKE THIS WORK FOR YOU

Your exercise selection should reflect your individual anthropometry, injury history, and equipment availability, not anyone else's programme. A tall lifter with long femurs may find high-bar squats uncomfortable and benefit more from a low-bar or front squat variation. A lifter with shoulder impingement history may need to avoid wide-grip barbell pressing and use dumbbells or neutral-grip machines instead. The best exercise for you is the one that loads the target muscle through a full range of motion without causing pain.

If you have limited equipment (home gym, minimal weights): Prioritise compound movements that load multiple muscle groups simultaneously. Single-leg variations (Bulgarian split squats, lunges) allow you to increase relative load without heavy weights. Calisthenics progressions (weighted pull-ups, dips, pistol squats) provide effective loading with minimal equipment. Your exercise selection is more constrained but the principles (progressive overload, full ROM, mechanical tension) remain the same.

If you have a specific muscle group that lags behind: Select exercises that target the lengthened position of that muscle and perform them first in your session when neural drive is highest. If your chest is lagging, dumbbell presses or flyes with a deep stretch should precede your heavier compound pressing. If your hamstrings are underdeveloped, Romanian deadlifts from the floor (full hamstring stretch) should precede leg curls.

The stability continuum in practice: Beginners benefit from machine-based work to learn the feeling of muscle tension before transitioning to free weights. Advanced lifters benefit from free-weight compounds for systemic load and machine isolations for targeted hypertrophy. Use both — they are complementary, not competing, training tools.

COACHING APPLICATION

Domain 1 Case Study: The Plateauing Novice

Scenario: A 24-year-old male client, 78 kg, has been training consistently for 6 months on a full-body programme three times per week. He made excellent linear progress for the first 4 months but has seen no strength increases on his main lifts (bench press, squat, row) for the past 6–8 weeks. He trains with moderate intensity and is unsure how to push past the plateau.

Assessment: (1) The linear progression model has exhausted its utility — he needs a phase transition to an intermediate progression model. (2) His training volume is approximately 8–10 sets per muscle group per week (likely at or slightly below MAV for his experience level). (3) He has never trained to failure or used intentional RPE targeting — his proximity to failure is likely RPE 5–6 (4+ RIR), meaning the tension stimulus is insufficient. (4) Exercise selection is limited to barbell compound movements without targeted isolation or stretch-mediated work.

Intervention: (1) Transition from linear progression to double progression in the 6–10 rep range. (2) Increase proximity to failure: target RPE 7–8 (2–3 RIR) for main work, with the final set of each exercise at RPE 9. (3) Add one isolation exercise per major muscle group targeting the lengthened position (incline dumbbell press for chest, RDL for hamstrings, lat pulldown with full stretch). (4) Increase chest volume to 12 sets/week and back volume to 14 sets/week. (5) Implement structured progression tracking: log every working set with load, reps, and RPE. (6) Add a scheduled deload every 6th week (50% volume, maintain intensity). Reassess after 8 weeks.

Key Takeaway: The novice plateau is caused by either insufficient stimulus (proximity to failure too low, volume too low) or insufficient progression structure. Transitioning from linear progression to double progression, ensuring sufficient proximity to failure, and adding stretch-mediated exercise selection typically restores progress.

Domain 2: Programming Variables & Periodization

6. Frequency — Per-Muscle Training Frequency and Recovery

6.1 Muscle Protein Synthesis Duration

Following a resistance training session, muscle protein synthesis (MPS) is elevated above baseline for approximately 24–48 hours in trained individuals. The magnitude and duration of this elevation depend on training volume, intensity, protein intake, and training status. The MPS elevation drives the adaptive response, meaning that the muscle is primed for growth during this window. Training a muscle group again while MPS is still elevated can compound the adaptive signal, but training it too frequently without sufficient recovery can accumulate fatigue without proportional adaptation.

6.2 Frequency Recommendations by Muscle Group

Table 6.1 Recommended Weekly Frequency by Muscle Group

MUSCLE GROUP	RECOMMENDED FREQUENCY (X/WEEK)	RATIONALE
Quadriceps	2-3	Large muscle group; rapid recovery; compound movements can be distributed
Hamstrings	2-3	Benefit from both hinge and leg curl stimuli; can handle higher frequency
Glutes	2-3	Large muscle group; often under-recovered if hit only 1x/week
Chest (pectorals)	2-3	2x/week minimum for growth; 3x in advanced volume accumulation
Back (lats, rhomboids, traps)	2-3	Large surface area; different movement angles per session
Shoulders (deltoids)	2-4	Smaller muscle; passive recovery in pressing; lateral raises tolerate high frequency
Biceps	2-3	Recover quickly; involved in pulling work; direct work can be 2x/week
Triceps	2-3	Recover quickly; involved in pressing; direct work 1-2x/week sufficient
Calves	3-6	High recovery capacity; benefit from frequent low-volume exposure
Abs / core	2-6	High recovery capacity; function improves with frequency

6.3 Split Structures and Frequency Mapping

Table 6.2 Split Types and Per-Muscle Frequency

SPLIT TYPE	TRAINING DAYS/WEEK	PER-MUSCLE FREQUENCY	VOLUME PER SESSION
Full body	3	3x/week	Low per muscle (4–6 sets)
Upper/Lower	4	2x/week	Moderate (8–12 sets)
PPL (Push/Pull/Legs)	6	2x/week	Moderate-high (10–14 sets)
Bro split (1x/week per muscle)	5-6	1x/week	High (15–20+ sets)
Torso/Limbs (Arnold)	4	Chest/back 2x, legs 2x, arms 2x	Moderate (8–12 sets)

CHAPTER SUMMARY

Muscle protein synthesis remains elevated for 24–48 hours post-training, driving the adaptive window. Most muscle groups respond optimally to 2–3 sessions per week. Smaller muscle groups (calves, abs, forearms) tolerate and benefit from higher frequencies. Split selection determines frequency: full-body = 3x/week, upper/lower = 2x/week, bro split = 1x/week. Higher frequency per muscle group allows lower per-session volume, reducing acute fatigue.

◈ MAKE THIS WORK FOR YOU

Your optimal training frequency depends on your recovery capacity, schedule, and training preferences, not just what the evidence says is optimal. Training a muscle group 2x/week produces equivalent hypertrophy to 3x/week when total weekly volume is equated — the key variable is volume per session, not frequency itself. If you can only train 3 days per week, a full-body split training every muscle 3x/week with lower per-session volume works as well as a 6-day PPL split with higher per-session volume.

If you have limited time (3 or fewer sessions per week): Full-body training is your most efficient option. You train every muscle group every session, so each muscle is stimulated 3x/week. The per-session volume is distributed (4–6 sets per muscle per session), keeping session duration reasonable while achieving 12–18 weekly sets per major muscle group.

If you train 4 days per week: Upper/Lower splits are the sweet spot for most people. Each muscle group is trained 2x/week with moderate per-session volume (8–12 sets). This split provides ample recovery between sessions while allowing sufficient volume accumulation per muscle group to drive progress.

If you prefer longer, less frequent sessions: A bro split (each muscle group 1x/week) can work, but you must accumulate sufficient volume in that single session (15–20+ sets per muscle) and accept that hypertrophy may be slightly suboptimal compared to 2x/week frequency for the same weekly volume. This is a valid preference-based choice if adherence is higher.

7. Rep Ranges and the Hypertrophy-Strength Continuum

7.1 The Overlap Model

Rep ranges exist on a continuum, not in discrete categories. All rep ranges between approximately 5 and 30 repetitions can produce hypertrophy, provided the set is taken to sufficient proximity to failure. The old dogma that "low reps build strength, high reps build endurance, and moderate reps build muscle" has been replaced by the overlap model: all rep ranges build muscle, but they do so through different mechanisms and with different efficiency. Low reps (1–5) primarily drive neural adaptation and myofibrillar hypertrophy. Moderate reps (6–15) balance mechanical tension and metabolic stress, producing the most efficient hypertrophy for most individuals. High reps (15+) create substantial metabolic stress and can produce hypertrophy but require more sets to reach equivalent fatigue of the targeted motor units.

7.2 Rep Range Adaptations

Table 7.1 Rep Range and Primary Adaptation

REP RANGE	%1RM	PRIMARY ADAPTATION	BEST FOR
1–3	90–100%	Maximal strength, neural drive, intramuscular coordination	Powerlifting peaking, strength-focused phases
3–5	85–90%	Strength with some hypertrophic stimulus	Strength accumulation, SBD-focused blocks
6–10	75–85%	Hypertrophy with strength carryover	Primary hypertrophy + strength hybrid work
10–15	65–75%	Hypertrophy (mechanical + metabolic)	Accessory work, isolation, volume accumulation
15–20	55–65%	Metabolic stress, muscular endurance	High-rep finishers, blood flow restriction adjuncts
20+	<55%	Muscular endurance, metabolite exposure	Specialised endurance, rehabilitation contexts

7.3 Muscle Fiber Type and Rep Range Optimization

Table 7.2 Fiber Type Distribution and Rep Range Recommendations

MUSCLE GROUP	APPROXIMATE TYPE I %	TYPE II %	REP RANGE RECOMMENDATION

Soleus (calves)	80–90%	10–20%	Higher reps (12-20+) for greater total tension time
Quadriceps (vastus lateralis)	40–50%	50–60%	Balanced (6-15); both ranges effective
Hamstrings	50–60%	40–50%	Moderate reps (8-15); stretch under load
Pectorals	40–50%	50–60%	Moderate reps (6-12)
Lats	45–55%	45–55%	Moderate reps (8-15)
Deltoids	40–60%	40–60%	Balanced (8-15); lateral raises benefit from higher reps
Biceps / Triceps	40–50%	50–60%	Moderate reps (8-15); respond to volume

7.4 Practical Rep Range Programming

Rather than prescribing a single rep range, the effective coach periodises rep ranges across a training cycle. A common pattern is: early mesocycle (6–10 reps, strength-accent), mid-mesocycle (8–15 reps, hypertrophy-accent), late mesocycle (10–20 reps, metabolic-accent or pre-deload). Different rep ranges can also be assigned to different exercises: compound lifts in the 5–10 range, isolation exercises in the 10–15 range, and finishers in the 15–20 range.

CHAPTER SUMMARY

All rep ranges from 5–30 can produce hypertrophy when taken to sufficient proximity to failure. Low reps (1–5) primarily build strength, moderate reps (6–15) are the hypertrophy sweet spot, and high reps (15+) add metabolic stress. Fiber type distribution varies by muscle group and may influence optimal rep selection. Periodising rep ranges across a mesocycle or assigning ranges by exercise type is more effective than using a single rep range year-round.

◈ MAKE THIS WORK FOR YOU

The rep range you emphasise should reflect your primary goal. If your main objective is hypertrophy, the majority of your work should be in the 6–15 rep range because it provides the best balance of mechanical tension, metabolic stress, and time efficiency. If your primary goal is strength (e.g., you compete in powerlifting), a larger proportion of your work will be in the 1–5 rep range, with hypertrophy work in the 6–12 range as a secondary block.

If your goal is pure hypertrophy: Use the 6–15 rep range for most exercises. Compound lifts can be in the 6–10 range (heavier, more mechanical tension), isolation exercises in the 10–15 range (more metabolic stress, better pump). Include some 15–20 rep work as finishers for muscle groups that benefit from high metabolite exposure (calves, lateral delts, forearms).

If your goal is strength with hypertrophy (the most common combination): Assign rep ranges by exercise. Your main compound movements (squat, bench, deadlift, row, overhead press) should stay in the 3–8 rep range to develop the neural coordination and skill component of strength. Your accessory and isolation work should be in the 6–15 rep range to drive hypertrophy that supports the main lifts. This hybrid approach is called "concurrent strength-hypertrophy programming" and works well for most non-competitive lifters.

If you feel you are "not responding" to a particular rep range: Before changing ranges, verify that your proximity to failure is adequate. It is common for lifters to leave too many reps in reserve in the 1–5 rep range (nervous system fears the heavy load) or in the 15+ rep range (the burn is uncomfortable). Ensure you are taking sets to within 1–3 RIR regardless of rep range, then assess.

8. Periodization Models for Hypertrophy

8.1 Why Periodize?

Periodization is the systematic planning of training variables over time to manage fatigue, maximise adaptation, and reduce injury risk. Without periodization, the lifter either stagnates (no progression) or accumulates excessive fatigue (overtraining). Periodization organises training into cycles: macrocycles (year+), mesocycles (4–12 weeks), and microcycles (weekly). For hypertrophy-focused athletes, the primary periodization decision is how to structure volume and intensity across the mesocycle.

8.2 Linear Periodization

Linear periodization involves increasing intensity (decreasing rep range) while decreasing volume over the course of a mesocycle. For example: weeks 1–3 (4 × 10–12), weeks 4–6 (4 x 8–10), weeks 7–9 (4 × 6–8), weeks 10–11 (5 × 4–6), week 12 deload. This model is intuitive and effective for beginners and intermediates. The advantage is clear progression structure; the disadvantage is that volume drops significantly in the later weeks, potentially reducing hypertrophic stimulus.

8.3 Daily Undulating Periodization (DUP)

DUP varies rep ranges within the same week, often across different training days. For example: Day 1 (heavy, 3–5 reps), Day 2 (moderate, 8–10 reps), Day 3 (light, 12–15 reps). DUP has been shown to produce superior strength gains compared to linear periodization in some studies, likely because it varies the stimulus frequently and avoids prolonged exposure to any single fatigue pattern. The disadvantage is that it requires more careful fatigue management and may be logistically demanding.

8.4 Block Periodization

Block periodization divides the mesocycle into distinct blocks, each with a specific focus. A common hypertrophy block structure: **Accumulation block** (4–6 weeks, high volume, moderate intensity, MEV+ to MAV volume), **Intensification block** (3–4 weeks, moderate volume, high intensity, MAV to MRV-), **Realization block** (1–2 weeks, low volume, high intensity, peaking). This structure allows the athlete to accumulate volume without excessive fatigue, then intensify the stimulus, then realise the adaptation in a peaking phase. Block periodization is preferred for advanced athletes who require longer accumulation phases to drive adaptation.

8.5 Periodization Model Comparison

Table 8.1 Periodization Model Summary

MODEL	STRUCTURE	BEST FOR	PROS	CONS
Linear	Decreasing reps, increasing intensity over mesocycle	Beginners, intermediates returning from break	Simple, clear progression, intuitive	Volume drops late mesocycle; limited stimulus variation
DUP	Varied rep ranges within the week	Intermediates, hypertrophy-focused athletes	Frequent variation, avoids accommodation	Logistically complex, fatigue management critical
Block	Accumulation → Intensification → Realization	Advanced athletes, specific peaking goals	Maximises adaptation per phase, manages fatigue	Longer mesocycle, requires patience

CHAPTER SUMMARY

Periodization systematically varies training variables to manage fatigue and maximise adaptation. Linear periodization (decreasing reps, increasing intensity over a mesocycle) is best for beginners and those returning from breaks. DUP (varying rep ranges within the week) is effective for hypertrophy-focused intermediates. Block periodization (accumulation → intensification → realization) is preferred for advanced athletes requiring longer, more structured phases.

◈ MAKE THIS WORK FOR YOU

Your periodization model should match your training age and preference. A beginner does not need DUP or block periodization — simple linear progression or linear periodization is sufficient and should be the default. An intermediate lifter benefits from DUP or simple block periodization (two blocks: accumulation then intensification). An advanced lifter (5+ years) likely needs structured block periodization with longer accumulation phases (6–12 weeks) to drive continued progress.

If you are a beginner (under 1 year): Use linear periodization or double progression. Your nervous system and muscles are adapting rapidly enough that complex periodization adds no benefit. Follow a simple structure: 4–6 weeks of moderate reps (8–12), then 4–6 weeks of slightly heavier work (6–8), with a deload between blocks. Keep it simple and consistent.

If you are an intermediate (1–3 years): DUP or two-block periodization works well. DUP is particularly effective if you enjoy variety within the week — heavy compound day, moderate volume day, light pump/metabolic day. Two-block periodization (accumulation block at 8–12 reps, then intensification block at 5–8 reps) provides a clear structure that balances volume and intensity.

If you are advanced (3+ years): Block periodization with three distinct phases is recommended. A 12-week mesocycle: accumulation (weeks 1–5, higher volume at 10–15 reps), intensification (weeks 6–9, moderate volume at 5–8 reps), and realisation (weeks 10–11, low volume at 3–5 reps), with week 12 as a deload. This structure allows the advanced lifter to accumulate enough volume to drive adaptation before transitioning to higher intensity work.

9. Powerlifting Periodization for Squat/Bench/Deadlift

9.1 The Conjugate Method

The conjugate method, popularised by Westside Barbell, involves rotating exercises and varying intensities simultaneously. Maximum effort (ME) work rotates through different competition lift variations (e.g., board press, deficit deadlift, box squat) to avoid accommodation. Dynamic effort (DE) work uses submaximal loads with explosive intent, typically 8–10 sets of 2–3 reps at 50–60% 1RM with short rest. Repetition effort (RE) or supplemental work builds muscle mass and addresses weak points. This system is highly effective for advanced powerlifters who have plateaued on standard linear or block periodization.

9.2 Block Periodization for Powerlifting

Table 9.1 Powerlifting Periodization Methods

METHOD	STRUCTURE	LIFTER LEVEL	DURATION
Linear (3-5 rep decrease)	12 weeks: 5x5 → 4x4 → 3x3 → peaking	Novice to early intermediate	12-16 weeks
Block (accumulation → intensification → peaking)	Week 1-6: volume (5x8-10 at 65-75%); Week 7-10: intensity (4x4-6 at 75-85%); Week 11-12: peaking (singles/ doubles at 85-95%)	Intermediate to advanced	12-16 weeks
Conjugate (ME/DE/RE)	Ongoing rotation of ME exercises; DE work 2x/week; RE supplemental	Advanced, plateaued	Continuous
Sheiko / high-frequency	3-4x/week per lift, moderate intensity, high volume	Intermediate to advanced	12-16 weeks

9.3 Peaking Protocols

A peaking block (typically 2–4 weeks) desensitises the lifter to heavy loads through exposure to near-maximal and maximal weights while reducing total fatigue. Key principles: **Reduce total volume** by 30–50% from the accumulation block. **Increase intensity** to 85–95%+ 1RM for competition lifts. **Eliminate or reduce accessory work** to the minimum necessary for maintenance. **Increase rest intervals** to 3–5 minutes to allow full neural recovery between sets. The final week typically includes a lighter recovery day 5–7 days before competition followed by complete rest or very light technique work 24–48 hours before the meet.

CHAPTER SUMMARY

Powerlifting periodization follows different structures depending on lifter level: linear progression (novice), block periodization (intermediate), or conjugate method (advanced, plateaued). Peaking protocols reduce volume by 30–50% while increasing intensity to 85–95%+ over 2–4 weeks. The sheiko method uses high-frequency, moderate-intensity training effective for lifters who respond well to volume accumulation.

◈ MAKE THIS WORK FOR YOU

Powerlifting periodization is specific to the sport — your programme should reflect whether you compete or simply train the three lifts for general strength. If you do not compete, you do not need conjugate periodization or a 12-week peaking block. A simple block structure (accumulation at 5–8 reps, then a short peaking phase before testing your max) is sufficient for non-competitive lifters who want to improve their squat, bench, and deadlift.

If you are a non-competitive lifter who trains the SBD lifts: Use a two-block approach. Block 1 (6–10 weeks): accumulate volume with 5–8 reps at RPE 7–8, focusing on technique and progressively overloading. Block 2 (2–4 weeks): test or push intensity with 2–5 reps at RPE 8–9, then deload. Repeat. This structure improves your competition lifts year-round without the complexity of formal peaking protocols.

If you are a competitive powerlifter: Your periodization should be planned around your competition calendar. Use block periodization (accumulation → intensification → peaking) for each meet cycle. The key to success is not the specific programme but consistent execution and managing fatigue so you arrive at the meet healthy and fresh. An incomplete training cycle with a forced deload due to accumulated fatigue is worse than a slightly simpler programme you can execute fully.

Weak point training: Identify your specific weakness within each lift (e.g., off-the-floor deadlifts, mid-range bench press, bottom of squat). Select targeted variations — deficit deadlifts, pause bench, front squats — and programme them as your primary or secondary movement during accumulation blocks. Direct weak point work is more effective than hoping the competition lift alone will fix the deficit.

10. Fatigue Management, Deload Design & Autoregulation

10.1 Sources of Fatigue

Fatigue in resistance training originates from three sources: **Metabolic fatigue** arises from the accumulation of metabolites (lactate, H+, Pi, ammonia) within the muscle during high-repetition, short-rest sets. This type of fatigue is acute, resolving within minutes to hours. **Neural fatigue** involves reduced central nervous system drive to the muscle, decreased motor unit recruitment, and altered cortical excitability. This resolves over 24–72 hours. **Systemic fatigue** encompasses hormonal (elevated cortisol, suppressed testosterone), inflammatory (cytokine elevation), and psychological (motivation, mood) components. Systemic fatigue accumulates over weeks and requires deload weeks to resolve fully.

10.2 Deload Protocols

Table 10.1 Deload Protocols by Context

PROTOCOL	DESCRIPTION	BEST FOR
Volume reduction (40-60%)	Keep intensity (load) the same but reduce sets by 40-60%	General maintenance deload; preserves neuromuscular readiness
Intensity reduction (60-70% 1RM)	Reduce loads significantly but keep volume and structure	Technique-focused deload; active recovery
Full rest deload	Complete rest from resistance training for 5-7 days	Systemic fatigue, high stress, illness recovery
Active recovery deload	Low-intensity cardio, mobility work, no resistance training	Mental break while maintaining movement quality

10.3 Deload Frequency Guidelines

Table 10.2 Deload Frequency by Training Phase

TRAINING PHASE	DELOAD FREQUENCY	RATIONALE
Novice (0-12 months)	Every 6-8 weeks	Lower training volumes; faster recovery; less systemic fatigue
Intermediate (1-3 years)	Every 5-7 weeks	Moderate volumes; periodisation naturally deloads

Advanced (3+ years)	Every 4-6 weeks	High volumes accumulate fatigue quickly; frequent deloads needed
Caloric deficit phase	Every 4-5 weeks	Reduced recovery capacity; more frequent deloads needed
High-volume accumulation	Every 4-6 weeks	Volume near MRV requires structured recovery

10.4 Autoregulation Methods

Autoregulation adjusts training variables based on the athlete's daily readiness. The three primary methods are: **RPE-based autoregulation** — prescribe a target RPE rather than a specific load. If the athlete feels fresh, they lift heavier at the same RPE; if fatigued, they lift lighter. **Repetition maximum (RM) testing** — the first working set of the day determines the load for subsequent sets. If the prescribed 8RM load moves for 10 reps, adjust up; if it only moves for 6 reps, adjust down. **Biofeedback autoregulation** — use heart rate variability (HRV), grip strength, or subjective readiness scores (1–10) to determine session intensity and volume. The simplest and most practical is RPE-based autoregulation: set the RPE target, let the athlete find the load.

When to Deload: Decision Process

1. **Schedule check:** Has it been 4–8 weeks since the last deload? If yes, schedule a deload.
2. **Performance decline:** Have strength or reps decreased for 2+ consecutive sessions for no clear reason? If yes, deload.
3. **Sleep/recovery decline:** Is sleep quality reduced? Resting HR elevated? Motivation low? If multiple indicators, deload.
4. **Training lag:** If training is suffering (joint pain, poor performance), deload regardless of schedule.
5. **Choose protocol:** Volume reduction is the default. Use intensity reduction if form needs work. Use full rest only if systemic fatigue is high or illness is present.

CHAPTER SUMMARY

Fatigue has metabolic, neural, and systemic sources. Deload protocols include volume reduction (40–60%, the default), intensity reduction, full rest, and active recovery. Deload frequency depends on training age (novice: 6–8 weeks; advanced: 4–6 weeks) and phase context (deficit phases need more frequent deloads). Autoregulation adjusts training to daily readiness, with RPE-based autoregulation being the most practical method.

◈ MAKE THIS WORK FOR YOU

Fatigue management and deload design are the most underrated elements of long-term progress. Most lifters either never deload (fearing they will lose progress) or deload too infrequently. Here is how to personalise this for your training context.

If you tend to undertrain (low volume, frequent inconsistency): You likely need fewer deloads — your fatigue accumulation is already low because your training volume is below MRV. Schedule a deload every 8–10 weeks as a mental reset and to check your progress. Use volume-reduction deloads (keep intensity, halve the sets) to maintain readiness.

If you tend to overtrain (high volume, high consistency): You are at risk of accumulating systemic fatigue that manifests as sleep disturbance, declining performance, and loss of motivation. Schedule a deload every 4–6 weeks without exception. Use full rest deloads (completely off from resistance training for 5–7 days) if you notice sleep or mood changes alongside performance decline. Volume-reduction deloads are the default otherwise.

RPE autoregulation for daily readiness: The single most practical skill you can develop is adjusting load to the day's readiness. Start each session with your first warm-up set and ask yourself: "Does this weight feel heavier, lighter, or the same as last week?" If heavier, reduce your working load by 5–10% while keeping the same RPE target. If lighter, increase it. This one skill prevents both undertraining on good days and overreaching on bad days.

Deload protocol quick reference: Volume reduction (keep intensity, drop sets by 40–60%) is the default for scheduled deloads. Full rest deloads are for systemic fatigue — when sleep, mood, or motivation is disrupted. Active recovery (walking, mobility, light cardio) is for when you need a mental break but want to stay in rhythm. Intensity reduction (drop loads to 60–70%, keep volume) is for technique-focused deloads.

11. Warm-Up, Potentiation & Session Structure

11.1 The RAMP Protocol

The RAMP protocol provides a structured warm-up framework: **Raise** — elevate body temperature, heart rate, and blood flow through 5–10 minutes of low-intensity cardio (cycling, rowing, incline walking). **Activate** — engage underactive or inhibited muscle groups through targeted activation drills (glute bridges, band pull-aparts, core bracing). **Mobilize** — take joints through their required range of motion for the upcoming session through dynamic stretching (leg swings, cat-camel, thoracic rotations). **Potentiate** — prepare the nervous system for heavy loads through progressive warm-up sets that approach working weight.

11.2 Warm-Up Set Progression

The warm-up set progression is critical for both performance and injury prevention. A general template for a compound lift: **Set 1**: 5–8 reps at 20–30% of working weight (empty bar or very light). **Set 2**: 3–5 reps at 50–60% of working weight. **Set 3**: 2–3 reps at 70–80% of working weight. **Set 4** (optional, heavy compound only): 1 rep at 85–90% of working weight. Rest between warm-up sets: 30–60 seconds for light sets, 60–90 seconds for heavier warm-ups. The total warm-up should not exceed 8–12 minutes for a single exercise to avoid pre-fatiguing the muscle.

11.3 Session Structure: Primary to Accessory

Table 11.1 Session Structure Components

COMPONENT	PURPOSE	DURATION	EXAMPLE
RAMP warm-up (general)	Increase core temp, activate, mobilize	8-12 min	5 min bike + glute bridges + leg swings
Warm-up sets (specific)	Potentiate the CNS, rehearse movement pattern	3-6 min	Progressive barbell warm-up toward working weight
Primary movement (compound)	Heaviest, most technically demanding lift of the session	15-25 min	Squat, bench press, deadlift, or main variation
Secondary movement (compound)	Second compound movement in different plane	12-18 min	RDL after squats, incline press after bench

Accessory lifts (isolation)	Targeted hypertrophy work, muscle-specific	15–25 min	Leg curls, lateral raises, curls, pushdowns
Cool-down / mobility	Restore resting state, improve flexibility	5–10 min	Light stretching, foam rolling targeted areas

11.4 Exercise Sequencing Principles

Exercise order within a session follows two principles. **The fatigue principle:** exercises requiring the highest neural demand and technique precision should be performed first, when the CNS is fresh. Multi-joint, heavy compound lifts (squat, bench, deadlift) always precede single-joint isolation. **The potentiation principle:** heavier, lower-rep sets can potentiate subsequent higher-rep work through post-activation potentiation (PAP). For example, heavy squats (3 reps at 85%) may enhance subsequent volume work (4 × 10 at 65%) by improving motor unit recruitment.

CHAPTER SUMMARY

The RAMP protocol (Raise, Activate, Mobilize, Potentiate) provides a structured warm-up framework. Warm-up sets progress from light to approximately 70–90% of working weight over 3–4 sets. Session structure flows from compound to isolation, heavy to light, complex to simple. Primary movements are performed first when the CNS is fresh. Post-activation potentiation can be exploited by sequencing heavier work before volume work.

◈ MAKE THIS WORK FOR YOU

Your warm-up and session structure should reflect your training environment, available time, and individual mobility needs. A 25-minute warm-up is not feasible for someone training before work in a crowded gym, and a 5-minute warm-up is insufficient for someone with a history of mobility restrictions.

If you train early morning (fasted or with minimal time): Extend your RAMP warm-up to 10–15 minutes. Your core temperature is lower, joints are stiffer, and the nervous system is less primed. Prioritise Raise (5 minutes of cardio — jumping jacks, bike, or brisk walk) and Potentiate (4–6 warm-up sets for your first compound lift, not 2–3). Do not skip activation — glute bridges and band pull-aparts are non-negotiable for early morning lifters.

If you train in the afternoon/evening: Your core temperature is naturally higher and movement quality is better. A 5–8 minute RAMP warm-up is sufficient. Focus on Mobilize for any restricted joints (hips, shoulders, thoracic spine) and Potentiate with 3–4 warm-up sets. Your activation work can be minimal if you have no known inhibition patterns.

If you have limited gym time (30–40 minute sessions): Collapse your warm-up into the first exercise. Do 3–4 warm-up sets for your primary compound movement while gradually increasing load. Skip general Raise and activation — the compound movement itself will raise core temperature and activate relevant muscles. Structure: 3 warm-up sets (1–2 minutes) → 3–4 working sets of primary compound (10–12 mins) → 2–3 sets of secondary compound (6–8 mins) → 2–3 sets of one isolation (5–6 mins) → done.

Exercise sequencing for structural balance: Always put your priority movement first. If chest is your weak point, bench before overhead press. If glutes are lagging, hip thrusts before squats. If you have no specific weak point, follow the standard order: compound → compound → isolation, heaviest to lightest. This ensures your freshest neural state is applied to the most demanding movement.

COACHING APPLICATION

Domain 2 Case Study: Intermediate Fatigue Management

Scenario: A 30-year-old male client, 88 kg, has been training consistently for 3 years on an Upper/Lower split, 4 days per week. He is 8 weeks into a hypertrophy block with progressive volume. He reports declining performance in the last 2 weeks — his working sets are losing 1–2 reps, his sleep quality has declined, and he feels "heavy" and unmotivated to train.

Assessment: (1) Systemic fatigue has accumulated over 8 weeks of progressive volume. (2) He has not taken a deload in 10 weeks. (3) Current volume is approximately 16–18 sets per muscle group per week, approaching his MRV. (4) The performance decline and sleep disturbance are classic signs that recovery capacity has been exceeded.

Intervention: (1) Immediate deload: 1 week at 50% volume (reduce sets by half, keep intensity the same). (2) Post-deload: reset volume to 70% of pre-deload levels (approximately 11–13 sets per muscle group). (3) Implement scheduled deloads every 6 weeks regardless of subjective feeling. (4) Switch from linear block to DUP to vary stimulus within the week: Day 1 (6–8 reps, RPE 8), Day 2 (10–12 reps, RPE 8), Day 3 (8–10 reps, RPE 7), Day 4 (12–15 reps, RPE 8). (5) Introduce RPE-based autoregulation: if the prescribed RPE feels 1+ point harder than expected on warm-up sets, reduce load accordingly.

Key Takeaway: Fatigue management is the primary constraint on long-term progress. Deloads should be scheduled pre-emptively, not reactively. DUP can distribute fatigue across the week more evenly than linear block periodization. RPE-based autoregulation prevents individual sessions from derailing the block when daily readiness varies.

Domain 3: Applied Split Design

12. Split Selection Principles

12.1 Decision Factors

Selecting a training split requires balancing training frequency, volume capacity, schedule constraints, and individual recovery capacity. No single split is universally superior; each represents a different trade-off between frequency, volume per session, and recovery demand. The optimal split is the one that allows the client to accumulate sufficient volume at the required intensity while fitting their schedule and respecting their recovery capacity.

Table 12.1 Split Selection by Available Days

AVAILABLE DAYS/WEEK	SUITABLE SPLIT OPTIONS	KEY CONSIDERATIONS
2	Full body (2x); Upper/Lower (if 2 consecutive days available)	Volume per session must be high; recovery between sessions critical
3	Full body (3x); Push/Pull/Legs (1x each); Upper/Lower with one rotation	Full body preferred for balanced development; frequency adequate
4	Upper/Lower (2x each); Torso/Limbs; Push/Pull/Legs + 1 full body	Upper/Lower is the most efficient for balanced hypertrophy
5	Upper/Lower + accessories; Push/Pull/Legs + Upper/Lower hybrid; Bro split	Allows for extra arm/shoulder volume; recovery becomes the limit
6	Push/Pull/Legs (2x each); Bro split (2 on/1 off cycle)	High frequency; volume must be managed to avoid exceeding MRV

12.2 Goal-Driven Architecture

The training goal determines the split architecture's emphasis. For **general hypertrophy**, balanced volume across all muscle groups 2x/week (Upper/Lower or PPL) is optimal. For **strength-focused hypertrophy**, the split should arrange sessions so that SBD lifts are performed fresh, with hypertrophy work as supplemental. For **weak point training**, the lagging muscle group is trained first in the session or given an additional weekly session. For **powerlifting peaking**, accessory volume drops and competition lifts take priority, which shifts split structure toward an Upper/Lower or competition-day split.

Key Distinction

Split design is not about which muscles are trained on which day. It is about **managing recovery** and **optimising performance** across the training week. The best split is the one the client can adhere to consistently.

CHAPTER SUMMARY

Split selection balances frequency, volume capacity, schedule constraints, and recovery. Full body (3x) is optimal for lower training days; Upper/Lower (4x) is the efficiency sweet spot; PPL (6x) maximises frequency. The training goal determines emphasis: balanced for general hypertrophy, SBD-prioritised for strength, and targeted for weak point training. The best split is one the client can adhere to consistently.

◈ MAKE THIS WORK FOR YOU

Your split selection should reflect your real schedule, not your ideal schedule. The most common mistake is selecting a 6-day PPL split when your life realistically allows 3–4 sessions per week. If you miss sessions, the frequency advantage disappears and you end up with inconsistent training. Choose the split you can execute consistently at least 90% of the time, then optimise from there.

If you have only 2–3 days per week: Full body is your best option. Each session trains every major muscle group, giving you 2–3x frequency per muscle group even on limited days. Keep sessions to 45–60 minutes with 4–6 sets per muscle group. Do not try to fit a 4-day split into 3 days by cramming — you will either overreach or under-recover.

If you have 4 days per week: Upper/Lower is the efficiency sweet spot. Two upper days and two lower days provide 2x frequency with sufficient per-session volume. This split works for most lifters who want balanced hypertrophy. It fits naturally into a Monday/Tuesday/Thursday/Friday schedule with weekends off.

If you have 5–6 days per week: You have options. PPL (6x) maximises frequency and is excellent for hypertrophy-focused lifters who enjoy daily training. Torso/Limbs (4-day rotation) provides great recovery by grouping synergistic movements. If you choose PPL, be honest about whether you can sustain 6 sessions per week for 12+ weeks — many lifters burn out by week 6.

The "weak point" modifier: If a specific muscle group is lagging, you do not need a new split. Simply adjust the split you already have: train the weak point first in its session, add 1–2 extra sets per week, or add a dedicated weak-point day. Changing splits is rarely the solution for a lagging muscle group.

13. Full-Body Split Templates

13.1 Full-Body Training Principles

Full-body training involves training all major muscle groups in each session, typically 3 times per week with at least 48 hours between sessions. This split maximises per-muscle frequency (3x/week) while minimising per-session volume (4–8 sets per muscle group). Full-body training is ideal for novices who benefit from frequent technique practice, individuals with limited training days (2–3 per week), and those who prefer shorter, more frequent sessions. The key challenge is managing systemic fatigue: because every session taxes the entire body, recovery between sessions must be prioritised.

13.2 Full-Body Template Structure

A standard full-body template includes: one squat or hinge pattern, one horizontal or vertical press, one horizontal or vertical pull, and one isolation exercise per main muscle group. The specific exercise selection rotates across the three weekly sessions to provide variation while maintaining the full-body stimulus. For example: **Session A** (squat focus): squat, bench press, row, leg curl, lateral raise, triceps. **Session B** (hinge focus): deadlift variation, overhead press, pull-up, leg extension, bicep curl, calves. **Session C** (balanced): front squat or lunge, incline press, lat pulldown, RDL, lateral raise, triceps.

13.3 Volume Per Session Limits

Full-body training is volume-constrained by session duration and systemic fatigue. A typical full-body session should not exceed 6–9 exercises with 20–28 total working sets. Per muscle group, 4–8 sets per session is the typical range, totalling 12–18 sets per muscle group per week at 3x frequency. If volume per session exceeds approximately 30 sets, the session becomes too long and recovery between sessions is compromised.

CHAPTER SUMMARY

Full-body training (3x/week, 48+ hours between sessions) maximises frequency with moderate per-session volume. It is ideal for novices, those with limited days, and those who prefer shorter sessions. Per muscle group, 4–8 sets per session totalling 12–18 sets/week is typical. The sessions must be carefully sequenced to avoid excessive systemic fatigue. Exercise rotation across sessions provides variation while maintaining a full-body stimulus.

◈ MAKE THIS WORK FOR YOU

Full-body training is deceptively simple in concept but requires discipline in execution. Because every session taxes the entire body, recovery between sessions is the main constraint. Here is how to make full-body work for your context.

If you are a novice (under 1 year): Full-body 3x/week is ideal for you. Your nervous system adapts rapidly to the frequent exposure, and your recovery capacity exceeds your training stimulus. Rotate exercises across sessions (A/B/C) to get more movement practice. Keep sessions to 45–55 minutes with 5–7 exercises. Use a simple linear progression — add 2.5–5 kg each session on your primary compounds. You do not need periodization or complex exercise rotation at this stage.

If you have only 2 days per week: Full-body 2x/week can still be effective if you structure volume appropriately. You need higher per-session volume (6–8 sets per muscle group) to compensate for the lower frequency. Prioritise compound movements and minimise isolation. Accept that progress will be slower than 3x/week — this is a volume-frequency trade-off that cannot be fully compensated.

If you train full-body as an intermediate or advanced lifter: Full-body training at 3x/week requires careful fatigue management. Each session involves compound lifts that tax the entire CNS. Use a heavy/medium/light structure across the week: Session A (heavy, RPE 8–9), Session B (medium, RPE 6–7, more volume), Session C (light, RPE 5–6, technique focus). This wave structure distributes fatigue so you can train the entire body three times per week without accumulating excessive systemic fatigue.

Key sequencing rule: Allow at least 48 hours between full-body sessions. If you train Monday-Wednesday-Friday, you have adequate recovery. If you try Monday-Tuesday-Thursday, the Tuesday session will compromise your Thursday performance. Never train full-body on consecutive days unless you are using a heavy/light structure where one session is intentionally low intensity.

14. Upper/Lower Split Templates

14.1 The Upper/Lower Framework

The Upper/Lower split divides training into upper body sessions (chest, back, shoulders, arms) and lower body sessions (quadriceps, hamstrings, glutes, adductors, abductors, calves). The standard frequency is 2x per week for both upper and lower (4 sessions total), though variations with 3 upper/2 lower exist. This split provides an excellent balance between frequency (2x per muscle group), per-session volume (8–12 sets per muscle group), and recovery (48–72 hours between sessions for the same muscle group).

14.2 Client Template: Upper/Lower (4-Day)

The following template represents a real client Upper/Lower split designed for balanced hypertrophy:

Table 14.1 Upper 1 — Chest, Back, Delts, Arms

EXERCISE	SETS	TARGET	COACHING RATIONALE
Incline chest press	3	Upper chest, front delt	Emphasises clavicular head of pec major; stretch-mediated stimulus
Machine chest flies	3	Pectorals (stretch focus)	Constant tension at the stretched position
Lat pulldown	2	Lats (width)	Compound vertical pull; foundation for back width
T-bar row	2	Mid back (thickness)	Horizontal pull targeting rhomboids and mid traps
Kelso shrugs	1	Upper traps	Isolated trap targeting with scapular elevation
Shoulder press (machine/dumbbell)	1	Anterior and lateral delts	Compound overhead; single set sufficient given pressing volume
Lateral raises	2	Lateral delts	Critical for shoulder width; lateral delt often lags
Rear delt flies	1	Posterior delts	Balances shoulder development; postural benefit
Preacher curls	2	Biceps	Stabilised elbow flexion; targets long head stretch

Triceps pushdown	2	Triceps (lateral head)	Cable constant tension; targets lateral head

Analysis: Upper 1 prioritises pressing volume (6 chest sets total between press and flies) with balanced pulling (5 back sets). Shoulder work is split across anterior (from pressing and press), lateral (2 sets raises), and posterior (1 set flies). Arms receive moderate direct volume. Total chest volume (6 sets) is slightly below MAV for intermediate-advanced lifters; this would be supplemented by pressing in Upper 2.

Table 14.2 Lower 1 — Quadriceps, Hamstrings, Glutes, Adductors/Abductors, Calves

EXERCISE	SETS	TARGET	COACHING RATIONALE
Leg extensions	3	Quadriceps (rectus femoris, vastus medialis)	Isolation that targets the rectus femoris (crosses hip)
Leg curls	2	Hamstrings	Knee flexion isolation; targets long head
Adductors (machine)	2	Adductor magnus/longus/brevis	Often neglected; important for squat stability and leg development
Abductors (machine)	2	Gluteus medius/minimus, TFL	Hip stability; lateral glute development
Hip thrust	2	Gluteus maximus	Maximises glute activation through hip extension in shortened position
Calves raises (standing/seated)	2	Gastrocnemius, soleus	High recovery capacity; 2 sets are minimum effective volume

Analysis: Lower 1 is machine-dominant, isolating each leg muscle group. The absence of a free-weight compound (squat, leg press) is notable. Quadriceps receive 3 sets of extensions. Hamstrings: 2 sets of curls. The absence of a squat or hinge pattern means stretch-mediated loading of the quads and hamstrings is limited.

Table 14.3 Upper 2 — Chest, Back, Delts, Arms

EXERCISE	SETS	TARGET	COACHING RATIONALE

Incline chest press	2	Upper chest	Reduced volume from Upper 1 (3→2); fatigue management
Machine chest flies	3	Pectorals	Same volume as Upper 1; maintains total chest stimulus
Lat pulldown	2	Lats	Consistent volume across sessions
Seated row	2	Mid back	Variation from T-bar; different moment arm
Kelso shrugs	2	Upper traps	Increased volume; traps recover quickly
Vertical shrugs (dumbbell)	1	Upper traps	Additional trap stimulus
Lateral raises	2	Lateral delts	Consistent across sessions
Preacher curls	3	Biceps	Increased from Upper 1 (2→3); bicep volume emphasis
Triceps overhead extensions	3	Triceps (long head)	Changed exercise from pushdown; targets long head at stretched position

Analysis: Upper 2 shifts emphasis: biceps and triceps receive increased volume (3 sets each). The back pulls use a different rowing variation (seated row vs. T-bar). Triceps exercise changes from pushdown (lateral head focus) to overhead extension (long head focus). Chest pressing volume drops slightly (2 sets vs. 3) while flies remain at 3 sets, keeping total chest volume at 5 sets (vs. 6 for Upper 1, total 11 sets/week, within MAV range).

Table 14.4 Lower 2 — Quadriceps, Hamstrings, Glutes, Adductors/Abductors, Calves

EXERCISE	SETS	TARGET	COACHING RATIONALE
Leg extensions	2	Quadriceps	Reduced from Lower 1 (3→2); fatigue management across week
Leg curls	3	Hamstrings	Increased from Lower 1 (2→3); hamstring emphasis
Adductors (machine)	2	Adductors	Consistent volume
Abductors (machine)	2	Abductors, glute med	Consistent volume
Hip thrust	2	Glutes	Consistent volume; 4/week total glute stimulus
Calves raises	2	Calves	Consistent volume; 4/week total

Analysis: Weekly totals: Chest 11 sets, Back 12 sets, Shoulders 9 sets, Biceps 6 sets, Triceps 5 sets, Quads 5 sets, Hamstrings 5 sets. Back and chest volumes are in MAV range. Quad and hamstring volumes (5 sets/week each) are below typical MAV. The machine-isolation-dominant pattern for legs misses stretch-mediated stimulus of compound squats and RDLs.

CHAPTER SUMMARY

Upper/Lower splits provide 2x per muscle group frequency with sufficient per-session volume (8–12 sets). The client template reveals a hypertrophy programme with balanced upper body volume (chest 11, back 12, shoulders 9 sets/week) and lower volume below typical MAV (quads 5, hamstrings 5 sets/week). The machine-dominant lower body pattern is an intentional choice that targets isolation but misses stretch-mediated compound stimulus.

◈ MAKE THIS WORK FOR YOU

The Upper/Lower split is the most versatile and widely applicable template for intermediate lifters. Here is how to adapt the standard template to your specific needs.

If your upper body is lagging behind your lower body: Prioritise upper body development by adding an extra upper day (3 upper / 2 lower). Structure it as Upper 1 (heavy pressing focus), Upper 2 (volume pulling focus), Upper 3 (arms and shoulders emphasis). Keep lower body at 2 days with moderate volume to maintain rather than grow. This asymmetrical split lets you add upper body volume without extending session length beyond 60 minutes.

If your lower body is lagging: Add a third lower day (2 upper / 3 lower). The third lower day should focus on the specific lagging area (e.g., glute-focused lower day, quad-focused lower day, hamstring-focused lower day). You can keep upper body at 2 days with maintenance volume while adding targeted lower body volume.

If you are time-pressed (30–40 minute sessions): Upper/Lower works well because you can superset antagonistic pairs on upper days (chest press with row, shoulder press with pulldown) to cut session time by 30–40%. On lower days, superset quad and hamstring exercises (leg extension with leg curl). Keep rest to 60–90 seconds for accessories and 2–3 minutes for compounds. You can complete a productive Upper/Lower session in 35–40 minutes with strategic supersetting.

A note on lower body volume: The client template in this chapter shows lower body volume below MAV (5 sets per muscle group per week). If you run this exact template, consider adding at least one compound lower body exercise — leg press, squat, or RDL — to each lower day. This will bring quad and hamstring volume into the 10–16 set per week MAV range. The machine-only approach works for maintenance but is suboptimal for growth.

15. Push/Pull/Legs Split Templates

15.1 The PPL Framework

The Push/Pull/Legs (PPL) split divides the body into three functional categories: push (chest, shoulders, triceps), pull (back, biceps, rear delts, forearms), and legs (quadriceps, hamstrings, glutes, adductors, abductors, calves). At 6 sessions per week, each category is trained twice. The PPL split provides the highest training frequency (2x per muscle group) with moderate per-session volume (10–14 sets per muscle group). The key advantage is functional synergy: pushing and pulling movements naturally work together, and leg sessions are separated from upper body sessions, minimising systemic fatigue interference.

15.2 Client Template: PPL (6-Day)

Table 15.1 Push — Chest, Front Delt, Triceps

EXERCISE	SETS	TARGET	COACHING RATIONALE
Incline chest press	3	Upper chest, front delt	Primary chest compound; upper chest emphasis
Machine chest flies	3	Pectorals	Stretch-focused isolation; constant tension
Shoulder press	2	Anterior and lateral delts	Overhead compound; anterior delt stimulus
Lateral raises	2	Lateral delts	Shoulder width; lateral delt hypertrophy
Triceps pushdown	2	Triceps (lateral head)	Cable isolation; constant tension throughout ROM
Triceps overhead extensions	2	Triceps (long head)	Stretch at the lengthened position; targets long head

Analysis: Push day features 6 chest sets (3 press + 3 flies) which is at the lower end of MAV (10–14/week). Triceps receive 4 sets total across two exercises. Shoulder volume is 4 sets (2 press + 2 raises). Total push work = 14 sets.

Table 15.2 Pull — Back, Rear Delt, Biceps, Forearms

EXERCISE	SETS	TARGET	COACHING RATIONALE
Lat pulldown	2	Lats (width)	Vertical pull; lat width development

Seated row	2	Mid back	Horizontal pull; rhomboid and mid trap thickness
T-bar row	2	Mid back	Different horizontal pull angle; increased back volume
Kelso shrugs	1	Upper traps	Targeted trap isolation
Vertical shrugs (dumbbell)	1	Upper traps	Additional trap volume
Rear delt flies	2	Posterior delts	Often lagging; important for shoulder health and posture
Preacher curls	3	Biceps	Stabilised elbow flexion; 3 sets = primary arm volume
Forearm extensions / flection	1 each	Forearm extensors/flexors	Forearm development; grip strength carryover

Analysis: Pull day provides 10 back sets, 2 rear delt sets, 3 bicep sets, and 2 forearm sets. Total = 15 sets. Back volume is within MAV range.

Table 15.3 Legs — Quadriceps, Hamstrings, Glutes, Adductors/Abductors, Calves

EXERCISE	SETS	TARGET	COACHING RATIONALE
Leg extensions	2	Quadriceps	Isolation; targets rectus femoris
Leg curls	2	Hamstrings	Knee flexion isolation
Adductors (machine)	2	Adductors	Hip adduction; squat stability
Abductors (machine)	2	Abductors, glute medius	Hip abduction; lateral glute development
Hip thrust	2	Glutes	Hip extension; glute hypertrophy
Calves raises	2	Calves	Isolation; high frequency tolerated

Analysis: At 2x/week PPL, weekly totals: Quads 4 sets, Hamstrings 4 sets, Glutes 4 sets, Adductors 4 sets, Abductors 4 sets, Calves 4 sets. These volumes are below typical MAV ranges for all leg muscle groups. The absence of any compound leg movement means all leg work is machine isolation.

CHAPTER SUMMARY

PPL splits (6x/week) provide 2x frequency with moderate per-session volume. The client template shows push (14 sets) and pull (15 sets) volumes within useful ranges, but leg volume is significantly below MAV (4 sets per muscle group per week). The absence of compound lower body lifts is notable. Coaches should consider adding a squat or leg press variant to address quad volume and an RDL for hamstring stretch loading.

◈ MAKE THIS WORK FOR YOU

The PPL split demands significant time commitment (6 sessions per week) and is best suited to lifters who can sustain high frequency. Here is how to adapt it to your context.

If you run PPL at a commercial gym: Equipment availability will determine your exercise selection. On push day, if all bench presses are taken, have a backup plan: dumbbell bench press or machine chest press work as direct substitutes. On pull day, if the lat pulldown is occupied, do pull-ups or assisted pull-ups instead. On leg day, if the leg press queue is long, do goblet squats or walking lunges. A rigid exercise list that depends on specific equipment will lead to frustration and missed sessions. Prepare 2–3 substitutions per exercise.

If you have lagging legs on PPL: The standard PPL leg template in this chapter provides only 4 sets per muscle group per week — well below MAV. Fix this by adding compound leg work: add a squat variant (barbell, goblet, or leg press) at the start of each leg session, and an RDL or hip thrust as the second compound. This brings quad volume to 10–12 sets and hamstring/glute volume to 8–12 sets per week. The template's machine-only leg work is fine as accessory but insufficient as your primary leg stimulus.

If you struggle with elbow or shoulder fatigue on PPL: PPL can accumulate significant joint stress because push day loads the elbows and shoulders twice per week, and pull day also loads the elbows (through pulls). If you experience elbow or shoulder discomfort, reduce triceps isolation on push days (drop one triceps exercise), reduce biceps isolation on pull days, and consider running PPL in a Push/Legs/Pull/rest rotation (not Push/Pull/Legs) to add an extra recovery day between push and pull sessions.

Sustainability check: Before committing to PPL, ask yourself honestly: can you train 6 days per week for 12+ consecutive weeks? If your schedule, motivation, or recovery capacity does not support this, choose another split. An inconsistent 6-day PPL is worse than a consistent 4-day Upper/Lower.

16. Bro Split / Body-Part Split Templates

16.1 The Bro Split Framework

The bro split, or body-part split, dedicates each training session to one or two muscle groups, training each muscle group once per week. The classic version is: Chest, Back, Shoulders, Legs, Arms. The per-session volume is high (15–20+ sets per muscle group), and frequency is low (1x per week). While the bro split has fallen out of favour in evidence-based coaching due to the superiority of higher frequency (2x/week) for hypertrophy, it remains effective for advanced lifters who need very high per-session volume to reach MRV for a muscle group, individuals with schedule constraints that limit frequency, and clients who enjoy the focus and intensity of single-muscle-group sessions.

16.2 Torso/Limbs Split (Client Template)

The Torso/Limbs split (often called the Arnold split after Arnold Schwarzenegger's preferred structure) trains the torso (chest, back, shoulders) on one day and limbs (legs, arms) on another, typically in a 4-day cycle: Torso, Limbs, Rest, Repeat.

Table 16.1 Torso — Chest, Back, Delts

EXERCISE	SETS	TARGET	COACHING RATIONALE
Incline chest press	3	Upper chest, front delt	Primary press; upper chest emphasis
Machine chest flies	3	Pectorals	Stretch-focused chest isolation
Lat pulldown	2	Lats	Vertical pull
Seated row	1	Mid back	Horizontal pull; lower volume on this variation
T-bar row	2	Mid back	Primary horizontal pull; higher volume
Kelso shrugs	1	Upper traps	Trap isolation
Vertical shrugs	1	Upper traps	Additional trap volume
Shoulder press	1	Anterior/lateral delts	Overhead; single set sufficient given pressing volume
Lateral raises	2	Lateral delts	Shoulder width focus

Rear delt flies	1	Posterior delts	Posterior delt maintenance

Analysis: Torso session totals 18 sets across chest (6), back (8 including traps), and shoulders (4). When run 2x/week in the Torso/Limbs cycle, this provides 12 chest sets/week and 16 back sets/week, both within MAV. The key advantage is that torso sessions group synergistic pushing and pulling muscle groups.

Table 16.2 Limbs — Legs, Arms

EXERCISE	SETS	TARGET	COACHING RATIONALE
Leg extensions	3	Quadriceps	Quad isolation; 3 sets = moderate volume
Leg curls	3	Hamstrings	Hamstring isolation; 3 sets = moderate volume
Adductors (machine)	2	Adductors	Hip adduction for stability and leg development
Abductors (machine)	2	Abductors, glute medius	Hip abduction; lateral glute
Hip thrust	2	Glutes	Glute hypertrophy
Calves raises	2	Calves	Calf isolation
Preacher curls	2	Biceps	Stabilised bicep curl
Hammer curls	2	Brachialis, brachioradialis	Forearm/biceps brachialis development
Triceps pushdown	2	Triceps (lateral head)	Tricep isolation (lateral head focus)
Triceps overhead extensions	2	Triceps (long head)	Tricep isolation (long head stretch)

Analysis: The Limbs session totals 22 sets: legs (14), biceps (4), triceps (4). At 2x/week, this provides 28 leg sets/week, 8 bicep sets/week, and 8 tricep sets/week. Leg volume is within MAV range for each muscle group when distributed.

16.3 The Arnold Split (Chest & Back / Legs / Shoulders & Arms)

The Arnold split structures training around three sessions: **Day 1: Chest & Back** — all pressing and pulling work, capitalising on the antagonistic pairing for enhanced recovery. **Day 2: Legs** — full leg session. **Day 3: Shoulders & Arms** — dedicated delt work plus bicep and tricep isolation. This creates a 3-day rotation repeated cyclically, typically training 5–6 days per week. Each muscle group is trained 1.5–2x per week with high per-session volume and minimal fatigue interference between conflicting movement patterns.

CHAPTER SUMMARY

Bro splits (1x/week per muscle group) require high per-session volume (15–20+ sets). The Torso/Limbs split provides 2x/week frequency when run in a 4-day cycle. The Arnold split (Chest & Back / Legs / Shoulders & Arms) provides 1.5–2x frequency in a cyclical rotation, capitalising on antagonistic pairing for recovery.

◈ MAKE THIS WORK FOR YOU

Bro splits and body-part splits have a polarised reputation — many evidence-based coaches dismiss them entirely, while many experienced lifters have built impressive physiques using them. The truth depends entirely on your training age and goals.

If you are a beginner or intermediate: A standard bro split (each muscle group once per week) is suboptimal for hypertrophy compared to 2x/week frequency. Research consistently shows that 2x/week produces superior or equal growth to 1x/week at matched volumes. If you choose a bro split, you must accept slower progress. However, if adherence is significantly higher with a bro split (you genuinely enjoy training one muscle group per day and are more likely to show up), then the adherence benefit can outweigh the frequency disadvantage.

If you are an advanced lifter (3+ years): Bro splits can be effective because you need high per-session volume (15–20+ sets) to reach your MRV. Training each muscle group once per week allows you to accumulate that volume in a single session without worrying about recovery between sessions for the same muscle group. Many advanced bodybuilders use bro splits or the Arnold split successfully because their per-session volume requirements are so high that 2x/week would require unsustainable session lengths.

The Torso/Limbs alternative: The Torso/Limbs split (Arnold split) provides a middle ground. It pairs chest and back together (antagonistic pairing saves time), then shoulders and arms together, then legs. Run as a 4-day rotation (Torso, Limbs, Rest, Repeat), this gives you 2x frequency for each muscle group while keeping sessions focused and efficient. This is an excellent alternative for lifters who enjoy the body-part focus but want better frequency than a true bro split.

If you are time-efficient: The Arnold split's antagonistic pairing (chest/back, then shoulders/arms, then legs) allows supersetting of opposing movements, significantly reducing session time. A 45–50 minute session can cover 18–22 working sets when chest and back exercises are superset. This is the bro split variant I recommend to most clients who want body-part style training — it preserves the focus of a bro split while achieving 2x frequency.

17. Powerlifting-Specific Programming Templates

17.1 Powerlifting Programme Structure

Powerlifting programming is organised around the three competition lifts: squat, bench press, and deadlift. Accessory work targets weak points identified through lift-specific analysis. The structure typically follows an Upper/Lower or SBD-focused split where each competition lift gets 2–3 sessions per week (or 1 heavy + 1 variation session). The primary difference from hypertrophy programming is that the competition lifts are always trained fresh (first in the session), and accessory volume is secondary to SBD performance.

17.2 Standard Powerlifting Week Structure

Table 17.1 Typical Powerlifting Week (4-Day Upper/Lower)

DAY	FOCUS	COMPETITION LIFT	VARIATIONS / ACCESSORIES
Upper 1 (Heavy)	Bench press strength	Bench press (1-5 reps, 80-95%)	Close-grip bench, triceps, upper back, lateral raises
Lower 1 (Heavy)	Squat strength	Squat (1-5 reps, 80-95%)	Deadlift variation (RDL, deficit pull), quad accessory, hamstring curls
Upper 2 (Volume)	Bench press volume	Bench press (6-12 reps, 65-75%)	Overhead press, rows, pulldowns, arm work
Lower 2 (Speed)	Deadlift & squat speed	Deadlift (speed sets, 50-65%, 8x2) + Squat variation	Core, glute/hamstring accessory, calves

CHAPTER SUMMARY

Powerlifting programming centres on the three competition lifts. A typical 4-day Upper/Lower week has two upper sessions (one heavy bench, one volume bench) and two lower sessions (one heavy squat, one speed deadlift/squat variation). Programme selection should match the lifter's training age: linear for novices, periodised for intermediates, and conjugate/block for advanced lifters.

◈ MAKE THIS WORK FOR YOU

Powerlifting programming is highly specific to whether you compete or simply train the three lifts for general strength. Your approach should differ accordingly.

If you are a non-competitive lifter who wants a stronger bench, squat, and deadlift: You do not need formal powerlifting programming. A well-structured hypertrophy programme that includes the competition lifts as primary movements will increase your SBD numbers through general muscle growth. Simply ensure you squat, bench, and deadlift (or close variations) at least once per week as your primary compound movements. Add a second weekly session for each lift at a different rep range (e.g., heavy low reps one day, moderate volume the other). This "powerbuilding" approach is simpler, more sustainable, and often more effective for non-competitive lifters than formal powerlifting periodization.

If you compete or plan to compete: Your programme must be structured around meet timing. A standard approach is: 12–16 week mesocycle divided into accumulation (6–8 weeks of moderate reps at RPE 7–8), intensification (4–6 weeks of heavier low-rep work at RPE 8–9), and peaking (2–3 weeks, singles and heavy doubles at RPE 9–10), followed by a deload before meet day. The conjugate method is an option if you have plateaued, but block periodization is sufficient for most lifters.

Weak point identification for powerlifting: The most effective way to improve your total is to address specific weak points within each lift, not to run generic programmes. For squat: is the sticking point at the bottom (weak quads) or in the mid-range (weak glutes/hams)? For bench: is it off the chest (weak pegs) or at lockout (weak triceps)? For deadlift: is it off the floor (weak erectors, glutes) or at lockout (weak glutes, lats)? Select specific variations that target your weak point — pause squat, spoto press, deficit deadlift, pin press — and programme them as primary movements during accumulation blocks for 6–10 weeks.

Accessory selection matters: Your accessory work should support your competition lifts. If your deadlift lockout is weak, you need more glute and lat work, not more hamstring curls. If your bench is weak off the chest, more pec work (especially at longer muscle lengths) will help more than more triceps work. Every accessory should have a rationale tied to a specific weak point in one of the three lifts.

18. Exercise Substitution Logic

18.1 Substitution by Goal

Exercise substitution is guided by the goal of the training phase. For **strength-focused** phases, substitutions should preserve the competition lift pattern. Substitute close-grip bench for conventional bench (same pattern, reduced ROM), or pause squat for competition squat (same pattern, increased time under tension). For **hypertrophy-focused** phases, substitutions should preserve the stretch and tension profile of the target muscle.

18.2 Substitution by Joint-Friendliness

When a client presents with joint limitations: **Replace axial loading with supported loading** (barbell squat → leg press). **Replace free weights with machines** for controlled resistance profiles. **Reduce shoulder internal rotation** in pressing (barbell bench → neutral-grip dumbbell press). **Reduce spinal compression** in hip hinge (barbell deadlift → trap bar deadlift or hip thrust).

18.3 Substitution Reference Table

Table 18.1 Exercise Substitution by Target

TARGET MUSCLE	PRIMARY LIFT	ALTERNATIVE A	ALTERNATIVE B	REASON
Quadriceps	Barbell back squat	Front squat	Leg press / hack squat	Front squat reduces spinal shear; leg press eliminates axial loading
Hamstrings	Barbell RDL	Nordic curl	Leg curl (seated or lying)	Nordic curl targets eccentric; leg curl isolates without spinal load
Chest	Barbell bench press	Dumbbell bench press	Machine chest press	Dumbbell allows greater ROM; machine provides constant tension
Lats	Pull-up	Lat pulldown	Straight-arm pulldown	Pulldown adjusts load to ability
Mid back	Barbell row	Chest-supported row	Cable seated row	Chest-supported eliminates lower back fatigue

Glutes	Barbell hip thrust	Barbell glute bridge	Cable pull-through	Glute bridge for home setup
Lateral delts	Dumbbell lateral raise	Cable lateral raise	Machine lateral raise	Cable provides constant tension through ROM
Triceps	Close-grip bench press	Tricep pushdown	Overhead tricep extension	Pushdown for constant tension; overhead for long head stretch
Biceps	Barbell curl	Dumbbell incline curl	Cable curl	Incline curl stretches biceps; cable curl provides constant tension

CHAPTER SUMMARY

Exercise substitution should preserve the goal: strength phases maintain competition pattern; hypertrophy phases maintain stretch/tension profile. Joint-friendly substitutions replace axial with supported loading, free weights with machines, and internally rotated with neutral grip pressing.

◈ MAKE THIS WORK FOR YOU

Exercise substitution is one of the most practical skills you can develop as a lifter. Equipment availability, joint discomfort, and schedule constraints will inevitably require substitutions. Here is how to personalise the substitution logic.

If you train at a commercial gym with limited equipment: Build your programme around substitutions from the start. If your programme calls for a barbell back squat but the only squat rack is always taken, have back squats → goblet squats → leg press → Bulgarian split squats as a substitution chain ready. The key is preserving the target muscle group and stimulus (stretch, tension, load) rather than preserving the exercise. A leg press will not replicate the full-body coordination of a squat, but it will still grow your quads effectively.

If you have joint limitations (knees, shoulders, lower back): Use the substitution table proactively, not reactively. If you know your knees hurt during squats, substitute leg press + walking lunges before pain forces you to skip training. If your shoulders ache during flat bench, substitute incline dumbbell bench (less impingement risk) before the pain becomes chronic. Reactive substitutions (changing after injury) are necessary but proactive substitutions (changing before injury) keep you training consistently.

If you train at home with minimal equipment: You have the most constrained substitution options. Build your programme around the equipment you have, then accept that some substitutes are imperfect. A dumbbell RDL is not the same as a barbell RDL, but it still trains the hamstrings effectively. A floor press is not the same as a bench press, but it trains the chest and triceps through a slightly shorter ROM. Prioritise exercises that work with your equipment rather than forcing exercises that do not fit.

The rule of preservation: When substituting, preserve the stimulus in this order: (1) target muscle group, (2) lengthened position loading/stretch, (3) load intensity, (4) movement pattern. If you cannot load the same muscle group at the same stretch position with similar intensity, the substitution is poor. If the stimulus is preserved but the movement pattern differs, the substitution is fine.

19. Training Split Calculator & Decision Logic

19.1 Decision Framework

The training split calculator logic evaluates four inputs: **Available days per week**, **training goal** (hypertrophy, strength, or hybrid), **training age** (novice, intermediate, advanced), and **recovery capacity**.

Split Selection Decision Tree

1. **Available days:** How many days can the client consistently train? (2–6)
2. **If 2–3 days:** Full body. Volume: 4–6 sets/muscle/session.
3. **If 4 days:** Upper/Lower or Torso/Limbs. Volume: 8–12 sets/muscle/session.
4. **If 5 days:** Upper/Lower + accessories or PPL + Upper.
5. **If 6 days:** PPL (2x each). Volume: 10–14 sets/muscle/session.
6. **Goal modifier:** Strength → prioritise compounds; Hypertrophy → even distribution; Hybrid → block periodization.
7. **Recovery modifier:** Low recovery → reduce per-session volume by 20–30%, increase deload frequency.

19.2 Weekly Volume Calculation Logic

$$\text{Sets per week} = \text{Sets per session} \times \text{Sessions per week for that muscle group}$$

19.3 Split-Specific Constraints

Full body: limited by session duration and systemic fatigue (max 28–30 sets/session). **Upper/Lower:** upper body limited by shoulder fatigue, lower body by CNS fatigue. **PPL:** push limited by elbow/shoulder fatigue, pull by grip/elbow fatigue. **Bro split:** limited by session duration; allows highest per-session volume but lowest frequency.

CHAPTER SUMMARY

The split calculator logic evaluates days available, goal, training age, and recovery capacity. Per-session volume is derived by dividing target MAV by frequency. Each split has unique constraints that must be respected for sustainable programming.

◈ MAKE THIS WORK FOR YOU

The split calculator is a decision framework, not a prescription. Here is how to apply it to your specific situation.

Step 1: Be honest about your available days. Count the days you can actually train consistently, not the days you wish you could train. If your schedule allows 4 days but your motivation or energy levels mean you realistically train 3 days, use 3 days as your input. A split designed for 4 days that you only execute 3 days per week will produce inconsistent results because the volume distribution across sessions will be off.

Step 2: Account for session length. The calculator assumes 45–75 minute sessions. If your available sessions are only 30 minutes (lunch break training, early morning before work), you need higher frequency and lower per-session volume. For 30-minute sessions: full body 4–5x/week with 2–3 exercises per session works better than trying to fit an Upper/Lower session into 30 minutes.

Step 3: Consider recovery capacity modifiers. Your recovery capacity is affected by: sleep quality, nutrition, stress (work, life, family), and training history. If you sleep fewer than 6 hours per night on average, or your job is physically/mentally demanding, reduce per-session volume by 20–30% from the calculator's recommendation. You can always add volume later if recovery is adequate. Starting too high and crashing is the more common mistake.

Step 4: Test and adjust. No calculator can perfectly predict your response. Run the recommended split for 4–6 weeks, then assess: Are you recovering between sessions? Is performance stable or improving? Is your sleep quality maintained or declining? If the answers are positive, continue. If sleep or performance is declining, reduce volume by 10–20% or increase deload frequency.

The golden rule of split selection: The best split is the one you will actually follow consistently. A mediocre split executed at 90% adherence will outperform a perfect split executed at 50% adherence every time. Factor in your enjoyment, gym logistics, and lifestyle constraints when making the final decision.

COACHING APPLICATION

Domain 3 Case Study: Split Selection by Schedule

Scenario: A 35-year-old female client, 62 kg, has been training for 2 years on a full-body split, 3 days per week. Her work schedule has changed: she now has 5 available training days but only 45–50 minutes per session. She identifies her biceps and glutes as lagging areas.

Assessment: (1) 5 days available, 45–50 min per session. (2) Goal: body recomposition. (3) Training age: intermediate. (4) Per session time constraint limits volume to approximately 20–24 sets maximum. (5) Lagging areas: biceps and glutes.

Intervention: (1) Recommend Upper/Lower + accessories (4-day Upper/Lower + 1 day glute/arm focus). (2) Superset antagonistic pairs on time-pressed days. (3) Implement RPE-based autoregulation to stay within time constraints. (4) The extra day allows focused development of lagging glutes and biceps without compromising the main Upper/Lower structure.

Key Takeaway: When time per session is limited, split selection should prioritise frequency over per-session volume. Adding a dedicated weak-point day allows focused development without compromising the main structure.

Domain 4: Special Considerations & Coaching

20. Training Age & Program Design

20.1 The Training Age Spectrum

Training age refers to the cumulative number of years an individual has been engaged in structured resistance training. Training age determines physiological responsiveness, recovery capacity, and the complexity of programming required to drive adaptation. The three broad categories — novice, intermediate, and advanced — exist on a continuum, and the transition between them is gradual rather than discrete.

20.2 Novice (0–12 Months)

Novices are characterised by rapid adaptive responses to any resistance training stimulus. Neurological adaptations dominate early gains. Linear progression (adding load session-to-session) is effective because recovery capacity far exceeds the training stimulus.

Programming guidelines: full body 3x/week, compound lift focus, 8–12 reps per set, 6–10 sets per muscle group per week (near MEV is sufficient for growth), RPE 7–8 (2–3 RIR) to minimise excessive fatigue, minimal isolation work beyond 1–2 exercises per major muscle group. The primary coaching challenge is technique acquisition, not stimulus optimisation.

20.3 Intermediate (1–3 Years)

Intermediates have exhausted linear progression and require periodization to continue progressing. The adaptive response slows, requiring higher volume and more sophisticated programming structures. Programming guidelines: full body, Upper/Lower, or PPL split depending on schedule, 2–3x per muscle group per week, 10–16 sets per muscle group per week (MAV range), double progression or linear periodization, RPE 7–9 (1–3 RIR) with occasional failure exposure, structured deloads every 5–7 weeks, targeted isolation work added for lagging muscles. The coaching challenge shifts from technique to fatigue management and periodization.

20.4 Advanced (3+ Years)

Advanced lifters have approached their genetic ceiling. Progress is measured in months to years, and even maintenance requires significant training volume. Programming guidelines: 2–3x per muscle group per week, 12–20+ sets per muscle group per week (MAV to MRV), block periodization or DUP, RPE 8–10 for key working sets, frequent strategic failure exposure, deloads every 4–6 weeks minimum, weak point training prioritised over balanced development. The coaching challenge is managing the narrow margin between stimulus and overreaching.

Table 20.1 Programming by Training Age

VARIABLE	NOVICE (0–1 YR)	INTERMEDIATE (1–3 YRS)	ADVANCED (3+ YRS)
Frequency per muscle	2-3x/week	2-3x/week	2-3x/week
Volume per muscle/week	6-10 sets	10-16 sets	12-20+ sets
Progression model	Linear (session-to-session)	Double progression (week-to-week)	Periodized (mesocycle-to-mesocycle)
RPE range	7-8	7-9	8-10
Deload frequency	Every 6-8 weeks	Every 5-7 weeks	Every 4-6 weeks

Primary challenge	Technique acquisition	Fatigue management	Marginal gains, peak timing

CHAPTER SUMMARY

Training age determines programming approach: novices respond to linear progression with low volume, intermediates require periodization and higher volume, and advanced lifters need sophisticated periodization with high volume and tight fatigue management. The progression involves decreasing adaptive rate, increasing volume requirements, and shifting the coaching challenge from technique to fatigue to marginal gains.

◈ MAKE THIS WORK FOR YOU

Your training age determines how you should approach programming, but many lifters misidentify their training age. Training age is about years of **consistent, structured resistance training**, not total years since you first touched a weight. If you trained inconsistently for 5 years (long breaks, switching programmes every few weeks), you are still a novice in terms of adaptive potential.

If you are self-assessing as a novice (0–1 year of consistent training): Your primary goal should be technique and consistency, not the perfect programme. Run a simple full-body 3x/week programme with linear progression (add weight each session). Do not worry about periodization, MAV, RPE precision, or advanced fatigue management. Your body will adapt rapidly to almost any stimulus. The risk is not undertraining — it is doing something too complex that you cannot sustain. Keep it simple for your first 6–12 months.

If you are self-assessing as an intermediate (1–3 years): You are in the sweet spot where good programming makes a meaningful difference. You need periodization (double progression or block periodization), volume in the MAV range (10–16 sets per muscle group/week), and structured deloads (every 5–7 weeks). Your coaching challenge is fatigue management — learning to balance volume accumulation with recovery. Most intermediates fail to progress because they train too hard without deloading, not because their programme is wrong.

If you are self-assessing as advanced (3+ years): Progress will be slow (measured in months per PR), and your programming must be precise. You need high volume (12–20+ sets per muscle group/week), strategic proximity to failure (RPE 8–10), frequent deloads (every 4–6 weeks), and weak point prioritisation. The margin between productive stimulus and overreaching is narrow. Track everything — loads, RPE, sleep, recovery, and subjective readiness — so you can detect overreaching before it forces an unplanned break.

If you have been training inconsistently for years: Your chronological training age may be high (5+ years since first gym visit), but your adaptive training age may be much lower. Be honest with yourself: have you been consistent for at least 6 months at a time? If not, programme as a novice or early intermediate. You will be surprised how quickly you can progress with consistent, simple programming.

21. Injury Risk Management & Pain-Aware Programming

21.1 The Pain-Aware Framework

Pain-aware programming distinguishes between discomfort (normal training sensation), nuisance pain (irritating but not limiting), and danger pain (sharp, localised, or worsening). The coach's role is to educate the client on this distinction and to modify training around pain rather than through it. The principle of relative rest — continuing to train the unaffected regions while modifying load, ROM, or exercise selection for the affected area — preserves muscle mass and conditioning during rehabilitation.

21.2 The Three-Tier Pain Response System

Table 21.1 Pain Response Tiers

TIER	PAIN DESCRIPTION	ACTION	EXAMPLE
Green (safe)	Muscle burn, fatigue, mild DOMs	Continue training; no modification needed	Lactate burn in quadriceps during leg extensions
Yellow (nuisance)	Dull ache, mild joint discomfort, resolves during warm-up	Modify: reduce ROM, adjust load, substitute exercise	Mild patellar ache during squats that resolves after warm-up
Red (danger)	Sharp, stabbing, worsening during set, localised to joint/tendon	Stop the exercise immediately; substitute; refer if persistent	Sharp shoulder pain during overhead press that worsens with each rep

21.3 Common Injury Prevention Strategies

General principles for reducing injury risk in resistance training: **Progress volume slowly** — the 10% rule (do not increase weekly volume by more than 10%) provides a conservative ceiling. **Manage tissue capacity** — tendons adapt more slowly than muscle (6–12 weeks vs. 2–4 weeks); volume jumps that are safe for muscle may overload tendons. **Balance agonist-antagonist ratios** — adequate pulling volume relative to pressing (at least 1:1, preferably 1.2:1 or more) reduces shoulder pathology risk. **Include rotational and scapular work** — face pulls, band pull-aparts, and scapular push-ups maintain shoulder health under high pressing volume.

Critical: When to Refer Out

Coaches must recognise when pain exceeds their scope of practice. Red flags requiring medical referral include: sharp or stabbing pain, pain that persists beyond 2 weeks of modified training, neurological symptoms (numbness, tingling, weakness), joint swelling or instability, and pain accompanied by systemic symptoms (fever, unexplained weight loss, night sweats).

CHAPTER SUMMARY

The pain-aware framework uses a three-tier system (green/yellow/red) to guide exercise modification. Relative rest preserves muscle mass during injury recovery. Prevention strategies include slow volume progression, tendon-capacity awareness, balanced agonist-antagonist training, and scapular health maintenance. Recognise red-flag symptoms that require medical referral beyond the coach's scope.

◈ MAKE THIS WORK FOR YOU

Injury risk management is not just for lifters who are already injured — it is for everyone who wants to train long-term. Here is how to apply the pain-aware framework to your training.

If you have no current pain but want to prevent injuries: The most effective prevention strategies are (1) progress volume slowly — no more than 10–20% increase per week, (2) maintain a pulling-to-pressing ratio of at least 1:1 (ideally 1.2:1), (3) include scapular and rotator cuff maintenance work (face pulls, band pull-aparts, external rotations) 2–3x/week, and (4) deload proactively. Most injuries come from doing too much, too fast, for too long without a break. If you follow these four rules, you will dramatically reduce your injury risk.

If you currently have a nagging ache or pain: Use the three-tier system. Ask yourself: is this green (normal muscle burn, mild DOMS, no concern), yellow (dull ache that resolves during warm-up), or red (sharp, stabbing, localised pain that worsens)? For yellow: modify the specific exercise (reduce ROM, change grip, use a machine instead of free weight) and continue training everything else normally. For red: stop the aggravating exercise immediately, substitute a pain-free alternative, and if the pain persists beyond 5–7 days, see a medical professional.

The relative rest principle: If you have a shoulder issue on pressing, you do not need to stop training entirely. Continue training legs, pulling, and core. Substitute the aggravating press with a neutral-grip or machine alternative at a lower RPE. This maintains your muscle mass, conditioning, and training momentum while the injury resolves. Complete rest leads to detraining and often makes returning harder because you have lost conditioning.

Know when to refer out: As an individual lifter, you need to know when to see a professional. Red flags: sharp pain that does not resolve within a few days of modified training, pain that wakes you at night, numbness or tingling in the extremities, joint swelling or instability, and pain accompanied by fever or unexplained weight loss. Do not try to train through these symptoms — see a physiotherapist or sports medicine doctor.

22. Kinetic Chain Mapping

22.1 Closed vs. Open Chain

The kinetic chain describes the interconnected system of joints and muscles through which force is transmitted during movement. In a **closed chain** exercise, the distal segment is fixed (e.g., squat with feet on the ground). Force is transmitted through multiple joints sequentially, producing greater co-contraction and joint stability. In an **open chain** exercise, the distal segment moves freely (e.g., leg extension). These exercises isolate specific muscles more effectively but with less co-contraction and joint stability. A balanced programme includes both, with closed-chain exercises as the foundation for compound strength and open-chain exercises for targeted hypertrophy.

22.2 Regional Interdependence

Regional interdependence recognises that dysfunction in one region of the kinetic chain often manifests as symptoms in a different region. Common patterns include: limited hip mobility presenting as low back pain during squats, shoulder instability presenting as elbow pain during pressing, and ankle restriction presenting as knee valgus during squats. Understanding these patterns allows the coach to address the root cause rather than the symptom.

22.3 Kinetic Chain Reference Table

Table 22.1 Kinetic Chain Mapping

JOINT/REGION	COMMON RESTRICTION	DOWNSTREAM COMPENSATION	AFFECTED LIFT	SCREENING TEST
Ankle dorsiflexion	<35° dorsiflexion	Knee valgus, heel rise in squat, forward lean	Squat (all variations), deadlift	Knee-to-wall test (>10 cm = adequate)
Hip flexion	Tight hip flexors, limited extension	Anterior pelvic tilt, lumbar extension, reduced glute activation	Squat depth, deadlift starting position	Thomas test; hip extension <10° = restricted
Hip external rotation	<30° passive rotation	Knee valgus, reduced squat depth	Squat, lunge, hip hinge	Seated hip rotation test

Thoracic extension	Inability to extend T-spine >20°	Lumbar extension compensation, forward head, shoulder impingement risk	Overhead press, squat (front rack), bench press	Wall slide test; T-spine rotation test (seated)
Shoulder internal rotation	<60° IR at 90° abduction	Scapular protraction, increased AC joint stress	Bench press (bottom position), overhead press	Supine shoulder IR test
Scapular stability	Winging, dyskinesis at rest or during movement	Reduced force transfer in pressing/pulling, increased impingement risk	All pressing and pulling variations	Scapular retraction test; wall angel test

Top-Down Screening Sequence

When assessing a movement limitation, screen from proximal to distal: (1) Core stability and breathing pattern, (2) Thoracic spine mobility, (3) Scapular control and shoulder mobility, (4) Hip mobility and pelvic control, (5) Knee tracking, (6) Ankle dorsiflexion. This sequence ensures that proximal restrictions (which often cause distal compensations) are identified first.

Kinetic Chain Screening Protocol

1. **Observe gait and stance:** Identify overt asymmetries or compensations.
2. **Screen overhead squat:** Watch for early heel rise (ankle), knee valgus (hip), lumbar flexion (hip/core), arm fall (T-spine/shoulder).
3. **Screen hip hinge:** Assess lumbar vs. hip contribution; restricted hip flexion presents as early lumbar rounding.
4. **Screen shoulder flexion/abduction:** Assess full ROM and quality of movement; note any scapular dyskinesis.
5. **Targeted assessment:** Based on findings above, perform specific ROM tests (Table 22.1) for restricted regions.
6. **Exercise modification:** Adjust exercise selection, stance width, grip width, or ROM to work within the client's available mobility.

CHAPTER SUMMARY

Kinetic chain mapping identifies how restrictions in one region cause compensations elsewhere. The top-down screening sequence (core → T-spine → shoulder → hip → knee → ankle) identifies root causes efficiently. Closed-chain exercises build foundational stability; open-chain exercises provide targeted isolation. The coach's role is to differentiate between mobility restrictions requiring specific intervention and stability limitations requiring motor control work.

◈ MAKE THIS WORK FOR YOU

Kinetic chain mapping is a diagnostic way to understand how your body compensates for mobility restrictions. Here is how to apply it to your own training.

If you struggle with squat depth: Do not assume it is an ankle issue. Use the top-down screening: start with core (can you brace without lower back arching?), then hip (test hip external rotation and flexion), then ankle (knee-to-wall test). Most squat depth issues originate from the hip (limited flexion or external rotation), not the ankle. If your hips are tight (positive Thomas test), spend 5–10 minutes daily on hip capsule stretches, couch stretch, and deep squat holds. If your ankles are the restriction (knee-to-wall distance <10 cm), perform daily ankle dorsiflexion drills with a band or slant board.

If you experience lower back pain during deadlifts: The most common cause is limited hip mobility, not a weak back. Test your hip external rotation (seated hip rotation test). If it is restricted (<30°), your lumbar spine compensates by rounding to achieve the deadlift start position. Fix the hip restriction with daily mobility work (banded hip distractions, 90/90 stretches, hip capsule CARs). Once hip mobility improves, the lower back pain often resolves without direct intervention.

If you experience shoulder pain during pressing: Screen your thoracic spine mobility first (seated T-spine rotation test). Limited T-spine extension forces the shoulder into excessive extension to achieve the bench press touch point, increasing impingement risk. If T-spine rotation is below 45°, add daily T-spine mobility work (foam roller extensions, open books, thoracic rotations in a half-kneeling position). A mobile T-spine unloads the shoulder joint.

Practical screening routine: Once per month, run through the top-down sequence: core brace test, seated T-spine rotation, wall angel, seated hip rotation, single-leg squat, knee-to-wall. Take 10 minutes. Note any restrictions. Address the most significant restriction with 5 minutes of daily mobility work for 2–4 weeks, then reassess. This proactive approach prevents many kinetic chain issues from becoming injuries.

23. Rehabilitation Protocols for Common Gym Injuries

Important Disclaimer

The protocols in this chapter are guidelines for training modification and return to exercise. They are **not** a substitute for professional medical diagnosis or physiotherapy. Any client with persistent, severe, or worsening pain should be referred to a qualified healthcare professional. Coaches operate within their scope of practice and should never diagnose injury.

23.1 Low Back Pain (LBP)

Mechanism: Often non-specific mechanical LBP from prolonged sitting, poor hip mobility, or excessive lumbar extension under load. Disc pathology is less common but more serious. **Red flags:** Radiating pain below the knee, numbness/tingling in the saddle region, bowel/bladder dysfunction, fever, unexplained weight loss. **Phased approach:** Phase 1 (protect, 1–2 weeks): remove spinal loading; substitute leg press for squat, chest-supported row for barbell row, trap bar deadlift for conventional. Phase 2 (load, 2–4 weeks): reintroduce axial loading at 50% volume; use goblet squats, RDLs with light loads. Phase 3 (return, 4–6 weeks): progressively return to full squat/deadlift variation at 70–80% of previous load over 3–4 weeks.

Table 23.1 Low Back Pain: Safe Exercise Regressions

AGGRAVATING EXERCISE	SAFE REGRESSION (PHASE 1)	INTERMEDIATE REGRESSION (PHASE 2)
Barbell back squat	Leg press / belt squat	Goblet squat / front squat (lighter)
Conventional deadlift	Trap bar deadlift	RDL (light, controlled eccentric)
Barbell row	Chest-supported row (machine)	Cable seated row (upright posture)
Overhead press (standing)	Seated dumbbell press (back supported)	Standing with core braced, lighter load

23.2 Shoulder Impingement

Mechanism: Subacromial narrowing causing mechanical compression of the supraspinatus tendon, subacromial bursa, or long head of the biceps tendon. **Red flags:** Night pain, painful arc (60–120° abduction), positive Neer or Hawkins-Kennedy tests. **Phased approach:** Phase 1 (protect, 1–2 weeks): eliminate overhead pressing and horizontal adduction past neutral;

substitute neutral-grip pressing, cable crossovers (low to high). Phase 2 (load, 2–4 weeks): reintroduce incline pressing at 60° (less impingement risk than flat bench), add face pulls and external rotation work. Phase 3 (return, 4–6 weeks): progressive return to flat bench and overhead press at reduced ROM (stop 2–3 inches above chest, no lockout).

23.3 Patellofemoral Pain (Runner's Knee)

Mechanism: Abnormal tracking of the patella in the femoral trochlear groove, often from weak VMO relative to lateral quadriceps structures. **Red flags:** Swelling, locking, giving way, pain behind the patella when sitting (movie sign). **Phased approach:** Phase 1 (protect, 1–2 weeks): eliminate deep squats and leg extensions; substitute leg press with limited ROM (0–45°), hip thrusts, step-ups (low step height). Phase 2 (load, 2–4 weeks): reintroduce partial ROM squats (to parallel), add terminal knee extensions (last 30°). Phase 3 (return, 4–6 weeks): progressively increase squat depth, reintroduce leg extensions at light weight.

23.4 Elbow Tendinopathy (Golfer's/ Tennis Elbow)

Mechanism: Overuse of the wrist flexors/extensors at the medial or lateral epicondyle. **Phased approach:** Phase 1 (protect, 1–2 weeks): eliminate direct forearm work; substitute neutral grip for pronated/supinated grip in pulling; use straps to reduce grip demand. Phase 2 (load, 2–4 weeks): introduce isometric holds (30–45 seconds) at mid-ROM; slow eccentrics (4–6 seconds) for wrist flexors/extensors. Phase 3 (return, 4–6 weeks): progressive return to full grip work; continue eccentric protocol 2–3x/week as maintenance.

23.5 Hamstring Strain

Mechanism: Eccentric overload during high-speed running or RDLs. **Phased approach:** Phase 1 (protect, 1–3 days): eliminate hamstring-dominant exercises; substitute hip extension work (hip thrusts, glute bridges). Phase 2 (load, 1–2 weeks): reintroduce RDLs at 30–50% load, 1–2 RIR; focus on slow eccentric (4–6 seconds). Phase 3 (return, 2–4 weeks): progressive loading to previous levels; add Nordic curls 2x/week for eccentric capacity.

23.6 DOMS vs. Injury

Table 23.2 DOMS vs. Injury: Differential Signs

FEATURE	DOMS	INJURY
Onset	12-24 hours post-training	Immediate or within minutes
Location	Diffuse, bilateral (if trained bilaterally)	Localised, often unilateral

Sensation	Dull ache, stiffness	Sharp, stabbing, or burning
Duration	24–72 hours	>5–7 days or worsening
Swelling	Mild or absent	Visible or palpable swelling/bruising
Symmetry	Bilateral (comparable body parts on both sides)	Unilateral (one side only)

Phased Return to Training After Injury

1. **Phase 1: Protect (1–14 days):** Remove the aggravating load. Maintain training for unaffected areas. Substitute with pain-free alternatives. Monitor: pain level (0–10), ROM, ADL function.
2. **Phase 2: Load introduction:** Begin light, pain-free loading of the affected structure. RPE 4–6 (4+ RIR). Slow concentric and eccentric tempos. Progress: increase load or volume every 4–7 days as tolerated.
3. **Phase 3: Progressive return:** Transition back to normal exercise selection at 60–70% of pre-injury load. Monitor for symptoms during and after sessions. If symptoms flare, reduce load by 20% and extend Phase 2 by 1–2 weeks.
4. **Re-injury prevention:** Identify the likely cause (excessive volume increase, poor form, mobility restriction). Address the root cause before returning to full training. Maintain an 80% load ceiling for 2–4 weeks after full return.

CHAPTER SUMMARY

Common gym injuries (low back, shoulder impingement, patellofemoral pain, elbow tendinopathy, hamstring strain) each require a phased approach: protect → load → return. Each phase has specific exercise substitutions and progression criteria. The DOMS vs. injury differential guide helps coaches and clients distinguish normal soreness from pathology. Coaches must never diagnose injuries; they modify training within the bounds of medical guidance.

◈ MAKE THIS WORK FOR YOU

Rehabilitation protocols are guidelines, not prescriptions. Every injury is individual, and your response to each phase will differ. Here is how to personalise the phased return approach.

If you have low back pain: Start with Phase 1 exercises immediately (leg press, chest-supported row, trap bar deadlift). Do not test your back with squats or conventional deadlifts until you have been pain-free during daily activities for at least 5–7 days. The most common mistake is returning too quickly — if Phase 1 feels easy for a few days and you jump back to squatting, the pain will return worse. Stay in Phase 1 for at least 1–2 weeks of complete pain-free training before attempting Phase 2 reintroductions.

If you have shoulder impingement: The single most effective change is switching to neutral-grip pressing (dumbbells with palms facing each other or a neutral-grip machine). This eliminates the internal rotation at the bottom of the press that irritates the supraspinatus. Most lifters can continue pressing with neutral grip without losing any chest or triceps stimulus. Only transition back to barbell bench after 4–6 weeks of pain-free neutral-grip pressing.

If you have patellofemoral pain (knee): Eliminate deep squats and leg extensions temporarily. Substitute leg press in a limited ROM (0–45° of knee flexion) and hip thrusts (which load the knees minimally). When reintroducing squats, start with box squats to a parallel height and gradually lower the box height over 3–5 weeks. This controlled ROM progression gives the patellar tendon time to adapt to the increased flexion angle.

General rule for the phased return: Proceed through the phases based on symptoms, not time. The timelines in this chapter are estimates. If Phase 1 symptoms persist beyond the estimated timeline, do not advance to Phase 2 — stay in Phase 1 and consider seeing a physiotherapist. If Phase 2 feels completely pain-free after a few sessions, you can advance to Phase 3 early. Listen to your body and adjust accordingly.

DOMS vs. injury reminder: If you are unsure whether you have DOMS or an injury, use this rule: if the discomfort is bilateral (both sides), diffuse, and peaks 24–48 hours after training, it is DOMS. If it is unilateral, localised, sharp, or immediate, it is likely injury. This is the most reliable self-assessment tool you have.

24. Female-Specific Training Considerations

24.1 Menstrual Cycle and Training Performance

The menstrual cycle creates predictable physiological fluctuations that affect training performance. Oestrogen and progesterone modulate neuromuscular function, substrate metabolism, thermoregulation, and recovery. Understanding these changes allows the coach to optimise training prescription without overcomplicating it. The evidence suggests that while performance fluctuations exist, individual variability is substantial and most female athletes do not need to periodize their training around the menstrual cycle unless they experience significant premenstrual symptoms.

Table 24.1 Training Considerations Across the Menstrual Cycle

PHASE	DAYS (28-DAY CYCLE)	PERFORMANCE CHANGES	TRAINING ADJUSTMENTS (OPTIONAL)
Early Follicular (menses)	1–5	Low oestrogen/progesterone; potentially reduced strength and power output; reduced neuromuscular fatigue resistance	Maintain normal training; may feel heavier; focus on effort-based autoregulation
Late Follicular	6–14	Rising oestrogen peaks before ovulation; improved neuromuscular performance; lower perceived exertion; peak strength/power window	Optimal window for heavy strength work, PR attempts, and high-intensity sessions
Luteal (early)	15–21	Rising progesterone; increased core temperature (0.3–0.5°C); reduced exercise capacity in hot conditions; increased protein catabolism	May require longer warm-up; slightly higher carbohydrate intake beneficial; monitor recovery
Luteal (late / PMS)	22–28	Highest progesterone; increased perceived exertion; reduced coordination; water retention; mood fluctuations; potential fatigue	Reduce expected output; prioritise technique; lower volume if needed; increase deload frequency if symptoms are severe

24.2 Hormonal Contraception

Hormonal contraception suppresses the natural menstrual cycle, eliminating the hormone fluctuations described above. Athletes on monophasic contraceptives experience a flat hormonal profile, meaning their training response is more consistent across the month. The anovulatory state may slightly reduce the late follicular strength peak, but for most athletes, the hormonal contraceptive state provides more predictable daily readiness. Coaches should ask about (but not pressure for) contraceptive status to inform training expectations.

24.3 Iron and Female Athletes

Menstruating females lose an average of 0.5–0.8 mg of iron per day through menstrual blood loss, with heavy menstruation adding 1–2 mg/day. This places female athletes at elevated risk of iron deficiency, which directly impairs aerobic performance and recovery. Annual ferritin screening is recommended. Subclinical iron deficiency (ferritin <30 ng/mL with normal haemoglobin) is more common than frank anaemia and still impairs performance. Supplementation should only follow testing to avoid iron overload.

CHAPTER SUMMARY

The menstrual cycle affects training performance through oestrogen and progesterone fluctuations. The late follicular phase (days 6–14) is typically the performance peak. However, most female athletes do not need strict cycle-based periodization. Hormonal contraception flattens the cycle's effects. Female athletes are at elevated risk for iron deficiency, requiring annual ferritin screening. The key coaching principle is individualised autoregulation rather than rigid cycle-based programming.

◈ MAKE THIS WORK FOR YOU

Female-specific training considerations are about understanding your individual response to hormonal fluctuations, not about rigidly restructuring your training around a calendar. Here is how to personalise this.

If you do not experience significant cycle-related symptoms: Do not change anything. The evidence does not support mandatory cycle-based periodization for women who feel consistent across the month. Train with the same programme structure year-round and use RPE-based autoregulation as your adjustment tool. The late follicular phase (days 6–14) will naturally be your best performance window — if you want to attempt PRs, schedule them there when convenient, but do not build your whole programme around it.

If you experience significant premenstrual symptoms (late luteal, days 22–28): This is where personalisation matters. In the week before menstruation, you may feel heavier, fatigued, less coordinated, and less motivated. Plan for this: (1) schedule this week's training at lower volume (reduce sets by 20–30%), (2) prioritise technique and lighter loads, (3) increase carbohydrate intake slightly to support glycogen, and (4) use RPE-based autoregulation — if the planned weight feels 1+ RPE harder than expected, reduce it. This is not "going easy" — it is managing your training around your biology so you can train consistently across the month.

If you are on hormonal contraception: Your hormone profile is relatively flat, which means your training response is more predictable across the month. You do not need to adjust training around a cycle. This is actually an advantage: you can programme normally with standard periodization and deloads. Just note that you may not experience the natural late-follicular strength peak — your performance will be consistent rather than cyclical.

Iron status check: If you menstruate and train regularly, ask your doctor for a ferritin test at least annually, especially if you experience unexplained fatigue, poor recovery, or performance plateaus despite adequate training and nutrition. Iron deficiency is common in female athletes and directly impairs recovery and performance. Do not supplement iron without testing — excess iron is harmful.

Strength training is safe and effective across the cycle: You can train heavy, go to failure, and build muscle in every phase of your cycle. The fluctuation is in daily readiness and perceived exertion, not in your capacity to adapt. Trust your autoregulation skills and train consistently.

25. Concurrent Training

25.1 The Interference Effect

Concurrent training refers to combining resistance training with endurance or cardiovascular training within the same programme. The interference effect describes the blunting of strength and hypertrophy adaptations when endurance training is added to a resistance training programme. The primary mechanisms are molecular (AMPK activation suppressing mTOR signalling), neuromuscular (residual fatigue from endurance work impairing motor unit recruitment during resistance training), and recovery competition (both modalities compete for limited recovery capacity). The interference effect is most pronounced when high-volume, high-intensity endurance work is performed in the same session as heavy resistance training.

25.2 Minimising Interference

Several strategies reduce the interference effect: **Separate sessions by at least 6 hours** (preferably 24+ hours). When same-session training is unavoidable, **resistance training first** produces better strength outcomes than cardio first. **Low-intensity steady-state (LISS) cardio** (<70% VO2max) produces substantially less interference than high-intensity interval training (HIIT). **Moderate endurance volume** (30–50 min, 3–4 sessions per week) can be accommodated without meaningful interference, provided protein intake (1.6–2.2 g/kg/day) is adequate. **Protein timing** — consuming 20–40 g protein before or after endurance sessions — may blunt the AMPK-mediated suppression of mTOR signalling.

Table 25.1 Concurrent Training Guidelines

SCENARIO	RECOMMENDATION
Strength + LISS (different sessions)	Minimal interference; separate by ≥6 hours
Strength + LISS (same session)	Resistance first, LISS second; moderate interference
Strength + HIIT (different days)	Moderate interference; prioritise recovery and protein intake
Strength + HIIT (same session)	Significant interference; avoid if possible; protein critical
Hypertrophy + endurance (any)	Moderate interference manageable with adequate nutrition

25.3 Practical Protocol

For a client combining resistance training with cardiovascular conditioning: schedule resistance sessions in the morning and LISS or metabolic conditioning in the afternoon (6+ hour separation). On resistance-only days, no additional interference concern exists. On cardio-only days (if separate from resistance days), no interference occurs. If both must be combined into one session, perform resistance training first (when CNS is fresh), then LISS (incline walk, cycling, rowing at low-moderate intensity). Avoid HIIT on resistance days unless the primary goal is cardio performance rather than hypertrophy.

CHAPTER SUMMARY

Concurrent training can produce interference via molecular, neuromuscular, and recovery-based mechanisms. LISS cardio creates minimal interference; HIIT creates more. Separate sessions by at least 6 hours or perform resistance training first. Moderate endurance volume (30–50 min, 3–4x/week) is compatible with hypertrophy goals when protein intake is adequate. The interference effect is real but manageable with proper programming and nutrition.

◈ MAKE THIS WORK FOR YOU

Combining strength training with cardio is one of the most common programming challenges. The right approach depends entirely on which goal is your priority and how much time you have.

If your priority is muscle growth, but you do cardio for health: Use LISS exclusively (incline walking, cycling, rowing at a conversational pace — 60–70% max HR). Three to four sessions of 20–40 minutes per week of LISS has essentially zero interference with hypertrophy when separated from resistance training by at least 6 hours (or done on separate days). Morning LISS before work + evening lifting is a perfect setup. Avoid HIIT unless you truly enjoy it — it produces the most interference and provides no hypertrophy benefit.

If your priority is cardio/endurance, but you want to maintain muscle: Keep resistance training 2–3x/week as your non-negotiable, and structure your cardio around it. Perform your hardest endurance sessions on non-lifting days. On lifting days, keep cardio to LISS or easy recovery work. Ensure protein intake at the higher end (1.8–2.2 g/kg/day) to protect muscle mass during the endurance volume. A 20–30 minute LISS session after lifting is fine; a 60-minute interval session after lifting is not.

If you do sport or recreation that requires both (e.g., football, MMA, military fitness): This is full concurrent training. Accept that neither strength nor cardio will reach its individual peak — you are optimising for both. Prioritise recovery aggressively: 7–9 hours of sleep, higher protein, scheduled deloads every 4–6 weeks. Train resistance first on combined days. Schedule your hardest sessions of each modality on separate days when possible. Consider block periodization across the year: blocks emphasising strength, blocks emphasising conditioning, rather than trying to max both simultaneously.

Same-session rule: If you must combine strength and cardio in one session, always lift first, cardio second. Never do hard intervals before lifting — the residual fatigue will wreck your lifting quality. After lifting, your choice: 15–20 minutes of LISS (minimal interference) or 10–15 minutes of metabolic conditioning (acceptable if your goal is conditioning, not max strength).

26. Home Gym / Minimal Equipment Programming

26.1 Programming Principles for Limited Equipment

Home gym programming requires adapting the principles of progressive overload to limited equipment availability. The key principles remain the same: mechanical tension, progressive overload, sufficient volume, and adequate frequency. The difference is that exercise variety is reduced, load progression is constrained by available weights, and certain movements require creative substitution. The most limiting factor is typically the lack of heavy compound loading and cable/machine isolation, which changes how you achieve tension and overload.

26.2 Equipment Categories and Programming Approach

Table 26.1 Home Gym Equipment and Programming Strategy

EQUIPMENT AVAILABLE	PROGRAMMING APPROACH	KEY LIMITATIONS
Dumbbells only (adjustable)	Full body or Upper/Lower; focus on single-leg work for loading; tempo and ROM as overload methods	Load ceiling on compound lifts; grip becomes limiting factor
Barbell + rack + bench	Full body or Upper/Lower; SBD-focused programming with barbell compounds	Expensive equipment; limited isolation; need plates for progression
Resistance bands + suspension trainer (TRX)	Full body with high repetition and tempo focus; time under tension replaces load progression	Limited load ceiling; progression via band thickness
Kettlebells	Full body with ballistic emphasis; swings, clean and press, Turkish get-ups	Limited progressive overload; strength ceiling low
Calisthenics (bodyweight only)	Full body with progression through leverage; push-up, pull-up, squat, lunge variations	Leg loading severely limited; lower body requires single-leg or plyometric progression

26.3 Progressive Overload Strategies for Limited Equipment

When load cannot be increased, alternative overload methods become critical: **Volume increase** — add sets and/or reps to increase total work. **Tempo manipulation** — slow the eccentric to 4–6 seconds, add a 1–2 second pause at the stretched position. **Decreased rest intervals** — reduce rest from 90 seconds to 45 seconds to increase density and metabolic stress. **Increased ROM** — deficit push-ups, deeper squats, full stretch at the bottom of each rep. **Unilateral work** — single-leg squats, single-arm pressing double the load on one limb. **Advanced variations** — decline push-ups, archer pull-ups, pistol squats, Bulgarian split squats.

CHAPTER SUMMARY

Home gym programming adapts standard principles to equipment limitations. Dumbbells, barbells, bands, kettlebells, and calisthenics each require distinct programming approaches. When load progression plateaus, alternative overload methods (volume, tempo, rest reduction, ROM, unilateral work) sustain adaptation. The most constrained home gym scenario is calisthenics for lower body, where single-leg work and plyometric progression are essential programming tools.

◈ MAKE THIS WORK FOR YOU

Home gym training is a constraint puzzle: you must achieve progressive overload with limited load options. Here is how to make your specific setup work.

If you have adjustable dumbbells only: This is the most common home setup and it can produce excellent results. Structure your training around unilateral and tempo-based loading: Bulgarian split squats (each leg carries a full dumbbell load), single-arm rows, and single-arm presses. When you hit the dumbbell load ceiling on a compound movement, switch to unilateral variations — a single 40 kg dumbbell Bulgarian split squat loads one leg with 40 kg, which is harder than both legs sharing 40 kg. Add a tempo protocol: 3–4 second eccentrics on every working set. This increases time under tension and makes moderate loads challenging. If you can, add one piece of equipment: a pull-up bar (transforms your back training) or a dip belt for weighted pull-ups/dips.

If you have a barbell and rack: You have the most flexible home setup. The main limitation is heavy loading for compound lifts — you will eventually outgrow the plates you own. Solution: invest in a few heavy plates incrementally (10 kg plates are the most valuable addition), or switch to higher-rep lower-load programming (10–15 rep ranges with shorter rest) which keeps moderate loads productive. Add a landmine attachment if possible — it enables a huge range of loaded movements (landmine presses, rows, squats) that are joint-friendly and require minimal space.

If you have bodyweight only: Your upper body can progress through leverage variations (decline push-ups, archer push-ups, pike push-ups, weighted-vest push-ups) and your back through pull-up progression (Australian rows → pull-ups → weighted pull-ups → archer pull-ups). The real challenge is legs — bodyweight squats, lunges, and jumps will not load the quads progressively. Use unilateral work aggressively: Bulgarian split squats, pistol squat progression, and weighted single-leg work (holding a backpack or child as load). Add a weight vest (20 kg) as the single most effective bodyweight-equipment upgrade — it re-enables progressive loading on everything.

If you travel frequently: Build a travel protocol: resistance bands (heavy set), a suspension trainer, or a door-frame pull-up bar. Do full-body sessions 3x/week with band-based compound movements (band squats, band press, band rows) at high reps (15–25) with slow tempos. Accept maintenance, not progress, during travel weeks. When you return to your gym, you will not have lost significant strength if you maintained volume and intensity with the bands.

The overload toolkit for limited equipment: When you cannot add weight, add: (1) more reps (up to 20–30 on isolation, 15–20 on compounds), (2) slower eccentrics (3–5 seconds), (3) pauses at the stretched position (2–3 seconds), (4) shorter rest (60 seconds), (5) unilateral variations, and (6) advanced leverage variations. Stack 2–3 of these at a time for continued progression.

27. Complete Training Assessment

A structured training assessment should be conducted at programme intake and reassessed after each training block (typically every 6–12 weeks). The assessment covers seven domains adapted from the Complete Nutrition Assessment framework:

27.1 Training History & Experience

- Total training age (years of consistent resistance training)
- Recent training history (last 12 months: split type, frequency, volume)
- Primary goal (hypertrophy, strength, power, endurance, general fitness)
- Competition history (powerlifting, bodybuilding, strongman, sports)
- Previous programmes tried and response to each

27.2 Movement Quality Assessment

Table 27.1 Movement Screening

ASSESSMENT	WHAT TO OBSERVE	RED FLAGS
Overhead squat (bodyweight)	Depth, knee tracking, torso angle, arm position	Early heel rise, knee valgus, lumbar rounding, arms fall forward
Hip hinge (RDL pattern)	Spinal position, hip excursion, knee bend	Lumbar flexion, excessive knee bend (squat pattern), limited hamstring stretch
Push-up (strict)	Scapular control, core stability, elbow path	Scapular winging, excessive flaring, lumbar sag
Pull-up or inverted row	Full ROM, scapular retraction, symmetry	Asymmetrical movement, partial ROM, lack of scapular control
Single-leg squat	Knee tracking, hip control, balance	Knee valgus, hip drop, loss of balance

27.3 Current Programme Audit

- Split type and weekly frequency per muscle group
- Current volume per muscle group per week (sets)
- Rep ranges used (primary and isolation)
- RPE/RIR targeting and proximity to failure
- Progression model (linear, double progression, periodized)

- Deload frequency and adherence
- Exercise selection (free weight vs. machine balance, ROM quality)

27.4 Recovery & Readiness Assessment

- Sleep quantity and quality (hours/night, sleep onset, awakenings)
- Stress levels (subjective 1–10; life and work stress factors)
- Training motivation (eager to train vs. dreading sessions)
- Joint health (any persistent aches, pains, or injuries)
- Illness frequency (colds, infections; elevated suggests overreaching)
- Heart rate variability (if tracked; declining trend = recovery concern)

27.5 Performance Tracking Audit

- Are working sets logged (exercise, load, reps, RPE)?
- What metrics are tracked? (scale weight, measurements, progress photos, performance)
- How frequently is progress assessed? (weekly, biweekly, monthly)
- Is there a defined progression plan or is training ad hoc?
- Are deloads scheduled or reactive?

27.6 Biofeedback & Subjective Experience

- How does the client feel during sessions? (energised vs. drained)
- Post-training recovery: appetite, sleep quality, next-day soreness
- Joint response to training (nagging pains, stiffness patterns)
- Training satisfaction (enjoyment vs. obligation)
- Workout consistency (missed sessions, shortened sessions, reduced effort)

27.7 Training Priority Matrix

Based on the assessment, prioritise interventions in order of impact:

1. **Consistency & adherence** — is the client training regularly? (first priority: nothing works without attendance)
2. **Progressive overload** — is load, volume, or quality of effort progressing? (second priority)
3. **Exercise quality** — are exercises well-selected with full ROM and appropriate stability? (third priority)
4. **Volume adequacy** — is volume in the MAV range for each muscle group? (fourth priority)
5. **Frequency & split optimisation** — is the split structure optimal for the goal? (fifth priority)

6. **Periodization & deload adherence** — is periodization managing fatigue effectively? (sixth priority)
7. **Recovery support** — sleep, stress, nutrition (supporting factors, not primary training variables)

CHAPTER SUMMARY

The 7-domain training assessment covers training history, movement quality, current programme audit, recovery, performance tracking, biofeedback, and a priority matrix. Consistency and progressive overload are the highest-priority intervention targets. The assessment should be conducted at intake and reassessed every 6–12 weeks. Movement screening identifies mobility and stability limitations before they become injuries.

◈ MAKE THIS WORK FOR YOU

The training assessment is not a one-time questionnaire — it is a review system you should run on yourself every training block (every 6–12 weeks). Here is how to apply it as a self-coached lifter.

If you are assessing yourself for the first time: Complete all seven domains honestly. The most valuable information will come from the Training History section and the Current Programme Audit. Write down: how many years of consistent training you have, what split you currently run, how many sets per muscle group per week, what RPE you typically use, and when you last deloaded. Most lifters discover they are training at volumes far from MAV or have not deloaded in months. The assessment surfaces the gaps that undermine progress.

If you feel stuck (no progress in 2+ months): Run the assessment with a specific focus on the priority matrix. In order: (1) Consistency — have you missed more than 10% of sessions? (2) Progressive overload — are loads or reps increasing? (3) Exercise quality — full ROM, good technique? (4) Volume adequacy — are you in MAV? (5) Frequency — is the split appropriate? (6) Periodization and deloads — are they scheduled? (7) Recovery — sleep, stress, nutrition. The first unmet priority is your problem. Most plateaus are consistency or volume problems, not split problems.

If you are returning to training after a break: Your assessment should focus on the deload principle: after 2+ weeks off, your training age has effectively reset. Do not resume at your previous volume and intensity. Resume at 50–60% of previous volume for 2–3 weeks, with RPE 6–7, then ramp back up. The movement screening section is especially valuable here — after a break, your mobility and stability will have declined. Spend extra time on the movement quality assessment before loading heavy.

If you train with a coach: The assessment framework gives you a vocabulary for your training review conversations. Track your answers over time (a simple notes file or spreadsheet) and bring the numbers to your review sessions. The best coaching relationships are built on accurate, honest self-assessment data.

Performance tracking minimum: If you track nothing else, log your working sets (exercise, load, reps, RPE) for your primary lifts. Ten minutes per week of logging gives you everything you need for the performance tracking audit and makes every future assessment faster and more accurate.

COACHING APPLICATION

Domain 4 Case Study: Returning from Low Back Injury

Scenario: A 34-year-old male client, 92 kg, has been powerlifting for 4 years (best lifts: squat 180 kg, bench 120 kg, deadlift 210 kg). He developed non-specific mechanical low back pain 6 weeks ago during a heavy deadlift session. He took 2 weeks off, but the pain returns when he tries to squat or deadlift. He wants to maintain his strength while recovering.

Assessment: (1) No red flags (no radicular symptoms, no bowel/bladder changes). (2) Pain is localised to the lumbar spine, worse with spinal flexion under load. (3) Hip mobility screening reveals <20° hip external rotation bilaterally and tight hip flexors (Thomas test positive). (4) The hip restriction forces lumbar extension compensation in the squat and lumbar rounding in the deadlift. (5) This is a kinetic chain issue: restricted hips are driving the lumbar overload.

Intervention: Phase 1 (protect, 2 weeks): substitute leg press for squat and trap bar deadlift for conventional at 40–50% of previous load, 2–3 RIR. Perform hip mobility drills daily (hip capsule stretches, banded hip distractions, couch stretch). Phase 2 (load, 3 weeks): reintroduce goblet squats and light RDLs at 50–60%. Add targeted glute med and core stability work. Phase 3 (return, 4 weeks): progressive return to squat (starting at 60% of previous max, adding 5 kg/week) and conventional deadlift (starting at 50%, adding 5 kg/week). Continue hip mobility as maintenance. Reassess pain levels after each session. If pain flares, reduce load by 20% and spend an additional week at Phase 2.

Key Takeaway: Low back pain in lifters is often a symptom of hip mobility restriction, not a primary spinal pathology. The phased return protocol (protect → load → return) maintains training momentum while allowing the underlying restriction to be addressed. The coach's role is to identify the kinetic chain cause (hip restriction) and prescribe appropriate mobility work alongside the training modification.

Appendix A: RPE / RIR Reference Chart

The Rating of Perceived Exertion (RPE) scale, adapted from Mike Tuchscherer's Reactive Training Systems, quantifies the intensity of a set based on the number of repetitions remaining in reserve (RIR). An RPE of 10 means no repetitions remain (failure); RPE 9 means one rep could have been completed; RPE 8 means two reps remain; and so forth. This autoregulation tool allows the lifter to adjust training load daily based on readiness rather than following a predetermined weight that may be too heavy or too light.

Using the RPE Scale

The RPE is assigned *after* the set is completed, not before. It reflects how many reps the lifter genuinely had left in the tank. Beginners frequently underestimate their RPE (thinking they had more reps left than they did). Video review and honest feedback from a coach accelerate accurate RPE estimation. The scale is most reliable at RPE 7–10; lower intensities (RPE 1–6) are better described as "warm-up" or "light" loading.

Table A.1 RPE / RIR Conversion

RPE	RIR (REPS IN RESERVE)	EFFORT DESCRIPTION	% OF 1RM (ESTIMATE)	SUGGESTED USE
10	0	Maximum effort; cannot complete another rep (absolute failure)	100%	Competition; planned max testing; occasional overload stimulus
9.5	0-1	Could not complete another rep, but the final rep was very slow	97-99%	Heavy singles/doubles; near-max strength work
9	1	One rep left in the tank; last rep was very challenging but controlled	93-96%	Heavy strength work; primary compound sets
8.5	1-2	Could have completed 1-2 more reps; hard but not maximal	90-93%	Strength-hypertrophy hybrid work
8	2	Two reps left; challenging but leaves clear reserve	85-90%	Primary hypertrophy work; most working sets

7.5	2–3	Moderately hard; 2-3 reps in reserve	82–87%	Volume accumulation; early mesocycle sets
7	3	Three reps left; controlled and smooth execution	78–83%	Volume work; technical practice; warm-up to work sets
6	4	Light-moderate; four or more reps in reserve	73–78%	Warm-up sets; technique work; deload sessions
5	5+	Light; five or more reps in reserve	68–73%	General warm-up; light recovery work
1-4	6+	Very light to minimal effort	<68%	Only for warm-up, mobility, or active recovery

A.1 Precision Rules

- **RPE only applies to working sets,** not warm-ups. Assign RPE only to the top 1–3 sets of an exercise.
- **Never assign RPE to a set performed to failure.** If you went to failure, that is RPE 10 by definition.
- **The %1RM estimates are for singles.** For multiple reps, the RPE reflects the total rep count, not just the weight. Example: 5 reps at RPE 9 means you completed 5 reps with 1 rep left. Five reps at RPE 10 means you completed 5 reps and could not get a 6th.
- **Account for the exercise's fatigue cost.** Squats and deadlifts accumulate more systemic fatigue per RPE unit than isolation work. An RPE 8 squat set feels significantly harder than an RPE 8 bicep curl set. Adjust expectations accordingly.
- **Train RPE estimation.** Periodically test RPE accuracy by performing a set to RPE 8, then asking the lifter to complete 2 more reps (to RPE 10). The gap between estimated and actual RPE reveals estimation error.

QUICK REFERENCE

RPE 7 = 3 RIR (volume work). RPE 8 = 2 RIR (primary hypertrophy). RPE 9 = 1 RIR (heavy strength). RPE 10 = 0 RIR (failure). Use the scale after each working set, not before. Adjust loads based on RPE: if RPE is too high, reduce weight 2.5–5%; if too low, increase weight 2.5–5% next session.

Appendix B: Deload Quick Reference

A deload is a planned reduction in training stress to dissipate accumulated fatigue, restore neuromuscular readiness, and reduce injury risk. Deloads are *proactive* (scheduled) rather than *reactive* (taken only when the lifter is already overtrained). This appendix consolidates the deload principles from Chapter 11 into a single quick-reference guide.

Table B.1 Deload Protocols by Training Age

TRAINING LEVEL	DELOAD FREQUENCY	VOLUME REDUCTION	INTENSITY REDUCTION	DURATION
Novice	Every 6-8 weeks	40-50% of normal	RPE 6-7 (3-4 RIR)	5-7 days
Intermediate	Every 5-7 weeks	50-60% of normal	RPE 5-6 (4-5 RIR)	5-7 days
Advanced	Every 4-6 weeks	50-70% of normal	RPE 4-6 (4-6 RIR)	7-10 days

Deload Decision Tree

1. **Is a scheduled deload due?** If yes, execute the planned deload. If not, check the signs below.
2. **Check accumulated fatigue signs:** Elevated RPE at normal loads, reduced motivation, sleep disruption, increased soreness, irritable mood. If 2+ signs present for 5–7 consecutive days, deload is indicated even if not scheduled.
3. **Check recent training:** Was there a volume or intensity spike in the last 2 weeks? If yes, a deload may be needed regardless of scheduled timing.
4. **Check external stress:** Increased work/life stress, reduced sleep, or illness. Even if training feels normal, an external stressor warrants a precautionary deload.
5. **If unsure, err on the side of deloading.** A week of light training costs nothing. Two weeks of accumulated fatigue can cost a month of progress.

B.1 Deload Execution Options

Table B.2 Deload Styles

DELOAD STYLE	METHOD	BEST FOR

Volume reduction (standard)	Same exercises, same intensity (RPE 6-7), halved sets	Most lifters; maintains motor patterns and stimulus
Intensity reduction	Same exercises, same volume, reduced load (50-60% of 1RM)	Lifters who need to maintain volume tolerance
Frequency reduction	Reduce sessions from 4-5x/week to 2-3x/week; full body	Lifters with high cumulative systemic fatigue
Active recovery	No resistance training; light cardio, mobility, stretching	Advanced lifters near competition; illness recovery

Common Deload Mistakes

- **Skipping deloads:** "I don't feel like I need one." Deloads are preventive, not reactive. Skipping them is the most common programming error and the leading cause of non-injury training plateaus.
- **Deloading too aggressively:** Reducing volume to <30% of normal creates a detraining response. The sweet spot is 40–60%.
- **Deloading with high intensity:** Keeping RPE high (8+) while dropping volume defeats the purpose. The deload is for fatigue dissipation, not stimulus maintenance.
- **Deloading by exercise substitution only:** Replacing every exercise with a new variation while keeping volume and intensity high is not a deload.

QUICK REFERENCE

Deload every 4–8 weeks depending on training age. Reduce volume 40–60%, keep RPE 5–7 (3–6 RIR), same exercises preferred. Watch for the four signs of accumulated fatigue. When in doubt, deload. A proactive deload costs one week; a reactive recovery break can cost a month.

Appendix C: Kinetic Chain Screening & Injury Quick Reference

This appendix consolidates the screening tests and injury protocols from Chapters 22 and 23 into a single coach-facing quick reference for field use.

C.1 Top-Down Screening Sequence

Table C.1 Screening Sequence

ORDER	REGION	QUICK TEST	PASS CRITERIA	COMMON COMPENSATION
1	Breathing & core	Observe breathing at rest; 360° expansion test; dead bug hold	Ribcage expands 3-dimensionally; core engaged without breath holding	Over-dominance of accessory breathing muscles; inability to brace without Valsalva
2	Thoracic spine	Seated T-spine rotation; wall slide test; half-kneeling T-spine extension	≥45° rotation each side; extension to neutral or slight extension	Lumbar extension to compensate for limited T-spine extension
3	Scapula & shoulder	Scapular retraction test; wall angel; empty can (supraspinatus)	Scapula sits flat on ribcage; full pain-free shoulder flexion to 180°	Scapular winging; forward shoulder posture; limited overhead flexion
4	Hip	Supine hip rotation; Thomas test; FABER test	≥30° external rotation; ≥10° hip extension in Thomas test	Lumbar extension (anterior tilt); knee valgus; reduced squat depth
5	Knee	Step-down test (from 20 cm box); single-leg squat	Knee tracks over 2nd toe; no valgus collapse; pelvis level	Knee valgus; hip drop; trunk lean to compensate

6	Ankle	Knee-to-wall test (weight-bearing dorsiflexion)	≥10 cm from wall (touching knee to wall without heel lift)	Early heel rise in squat; forward lean; knee valgus compensation

C.2 Injury Protocol Summary

Table C.2 Injury Quick Reference

INJURY TYPE	KEY RED FLAG	PHASE 1 SUBSTITUTE	PHASE 2 REINTRODUCTION	RETURN TIMELINE
Low back pain (non-specific mechanical)	Pain radiating below knee; saddle numbness; bowel/bladder changes	Leg press, trap bar deadlift, chest-supported row	Goblet squat, RDL (light), cable row	4–6 weeks
Shoulder impingement	Night pain; painful arc 60–120°; positive Neer/Hawkins-Kennedy	Neutral-grip press, cable crossover (low to high), face pulls	Incline press at 60°, external rotation work	4–6 weeks
Patellofemoral pain (runner's knee)	Swelling, locking, giving way, movie sign (pain after sitting)	Leg press (0–45°), hip thrust, step-up (low step)	Partial ROM squat (to parallel), terminal knee extension	4–6 weeks
Elbow tendinopathy (medial/lateral)	Point tenderness at epicondyle; pain with grip/wrist movement	Neutral grip all pulling; straps to reduce grip demand	Isometric holds at mid-ROM; slow eccentrics for wrist flexors/extensors	4–6 weeks
Hamstring strain	Immediate sharp posterior thigh pain; bruising; unable to weight-bear	Hip thrust, glute bridge (eliminate hamstring-dominant loading)	RDL at 30–50% load, slow eccentric (4–6s)	2–4 weeks

Coach's Scope Reminder

Coaches do not diagnose injuries. The protocols in this reference are training modification guidelines for clients who have already sought medical diagnosis or for whom the injury is clearly within the scope of mechanical overuse. Any red-flag symptom, unresolved pain beyond 2 weeks, or worsening progression requires immediate medical referral. Document all modifications and communications regarding injury management.

QUICK REFERENCE

The top-down screening sequence (core → T-spine → shoulder → hip → knee → ankle) systematically identifies kinetic chain restrictions. Common gym injuries follow a protect → load → return phased protocol. When in doubt about any symptom's seriousness, refer to a medical professional. The coach's primary role is training modification, not diagnosis.

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